MIL-M-55650B(EL)
28 November 1975
SUPERSEDING
MIL-M-55650A(EL)
30 January 1973

MILITARY SPECIFICATION
MULTIPLEXER TD-660( )/G

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Multiplexer TD-660( )/G which is a voice frequency processing unit providing 6 or 12 channels of communication. It contains transmit and receive sections. The transmit section provides 6 or 12 voice frequency channels for converting voice signals to Time Division Multiplex (TDM), Pulse Code Modulated (PCM) signals. The receive section converts PCM signals to voice signals simultaneously. (See 6.1 and 6.6).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Shipping, Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (for Carton Sealing)

FSC 5805

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MILITARY

MIL-P-116	Preservation Packing, Methods of
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Form, Expanded Bend)
MIL-P-26514	Polyurethane Foam, Rigid or Flexible for Packaging
MIL-M-38510	Microcircuits, General Specification For
MIL-B-43014	Boxes, Water Resistant Paperboard, Folding, Set-up and Metal Staved
MIL-I-46058	Insulation Compound, Electrical (For Coating Printed Circuit Assemblies)
MIL-T-55654	U TEMPEST Requirements and Test for Multiplexer TD-660( )/G (Classified)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	Definition of and Basic Requirements for Enclosures for Electronic and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-275	Printed Wiring for Electronic Equipment

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MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics Measurement of
MIL-STD-781	Reliability Test: Exponential Distribution
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-GL-323304	Gages for TD-660( )/G Multiplexer
SC-DL-524666	Panel Assembly, Dummy Test
SC-DL-524668	Panel Assembly, Coder
SM-B-524691	Panel Assembly 11A13, Voltage Reference
SM-B-317431	Test Requirement, Power Supply 11A33
SM-B-317432	Test Requirement, Panel Assy 11A33A1
SM-C-524874	Bracket, Mounting
SM-B-525082	Test Requirement, Multiplexer TD-660( )/G
SC-DL-525083	Multiplexer TD-660( )/G
SM-B-525090	Filter Assembly 11A19 Test Requirement
SM-B-525110	Test Requirement, Panel Assy 11A21 Talk Monitor
SM-B-525112	Test Requirement, Filter Assembly 11A18
SM-B-525115	Test Requirement, Panel Assembly 11A18A1 Buffer, PCM OUT ALT IN
SC-DL-525135	Cable Assembly, Power Electrical, CX-11727/G
SM-B-525214	Test Requirement, Panel Assembly 11A31A1, Generator, Transmitter Timing
SM-B-525218	Test Requirement, Filter Assembly 11A31, Generator, Transmitter Timing
SM-B-525219	Test Requirement, Panel Assembly, 11A31A2, Generator, Transmitter Timing
SM-B-525224	Test Requirement, Filter Assembly, 11A32, Generator, Receiver Timing
SM-B-525225	Test Requirement, Panel Assembly 11A32A1, Generator, Receiver Timing
SM-B-525230	Test Requirement, Panel Assembly 11A32A2, Generator, Receiver Timing

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SM-B-525237	Test Requirement, Panel Assembly 11A23, Transmit Timing
SM-B-525242	Test Requirement, Panel Assembly 11A24, Modem Timing and Output
SM-B-525247	Test Requirement, Panel Assembly 11A25, PAM Reshaper and Noise Generator
SM-B-525252	Test Requirement, Panel Assembly 11A26, Triple Modem
SM-B-525257	Test Requirement, Panel Assembly 11A27, Decoder, Receiver Input and Framing
SM-B-525262	Test Requirement, Panel Assembly 11A28, Receive Timing
SM-B-525267	Test Requirement, Panel Assembly 11A29, Fault Locator and Alarm Circuit
SC-DL-531012	Cable Assembly R.F. CG-1040( )/U
SC-DL-531014	Cable Assembly Special Purpose, Electrical CX-7870( )/TCC

PUBLICATIONS

DEPARTMENT OF THE ARMY

TM 11-5805-382-12	Operator and Organizational Maintenance Manual - Multiplexer TD-660( )/G

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and of Drawing and Data List SC-DL-525083, including the requirements for parts, materials and processes thereon.

3.3 Parts, materials and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

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3.4 Electrical characteristics. Multiplexer TD-660( )/G shall multiplex 6 or 12 voice frequency channels on a time division basis and modulate them using pulse code modulation. The multiplexed output signal shall be a binary full width pulse train at a 288 kilobit rate for 6 channel operation and a 576 kilobit rate for 12 channel operation. The receive section of the Multiplexer TD-660( )/G shall demodulate and demultiplex the binary signal and regenerate the voice frequency signals.

3.4.1 Channel characteristics.

3.4.1.1 Frequency response (see 4.8.10). Each channel shall transmit full modulation voice frequencies with deviations relative to 1100 Hertz (Hz) value as indicated.

FREQUENCY IN Hz	DEVIATIONS IN db
200 to below 300	-6.0 to +0.5
300 to 3450	-1.5 to +0.5
Above 3450 to 3500	-3.0 to +0.5
Above 3500 to 3640	-6.0 to +0.5

Each channel shall also transmit voice frequencies, at -14 dbm ± .1 db (4 wire) with deviations relative to 1100 Hz value of -6.0 to +0.5 db from 200 Hz to 3640 Hz.

3.4.1.2 Input Level 2/4 wire (see 4.8.21). An input 1100 Hz test tone level of -4±.1db, 4 wire shall cause full modulation of the audio channel. This signal level shall be used for the measurement of output level, signal-to-noise ratio and signal-to-noise plus cross-talk ratio.

3.4.1.3 Output level 2/4 wire (see 4.8.11). The output level for alignment purposes with full modulation at the modulator shall be +1.0 dbm ± .5, 2 wire and -4.0 dbm ± .5 db, 4 wire. The 2 wire output level shall be adjustable between -6.0 and +5.0 dbm.

3.4.1.4 Input-output linearity characteristics (see 4.8.22). The input-output characteristics of each channel at 1.1 Khz shall be linear within ± 20 millivolts RMS for input levels or -4.0 dbm to above -31.0 dbm and ± 5 millivolts RMS for input levels of -31.0 dbm to -35.0 dbm, 4 wire.

3.4.1.5 Audio level stability.

3.4.1.5.1 Stability at standard ambient conditions (see 4.8.31). Each channel output level shall not deviate more than ± 0.5 db from initial level in any 24 hour period when the equipment is operated at standard ambient environmental conditions, per MIL-STD-810.

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3.4.1.5.2 Stability at Temperature Extremes (see 4.6.1, 4.8.8). Each channel output level shall not deviate more than ± 2.0 db from initial level over the temperature range specified in 4.6.1.

3.4.1.6 Signal-to-Noise Radio (see 4.8.9). The signal-to-noise ratio of each channel shall exceed 55.0 db FIA weighting with two each multiplex equipments operated as a system. This value shall be met with the random noise source applied to all channels. For any one equipment, this ratio of 55.0 db shall be simultaneously met in at least ten of the channels in 12 channel operation; and four of the channels in 6 channel operation, and all channels shall exceed 49.0 db. The 12 channel signal-to-noise ratio requirement shall apply for any one equipment aligned for 24 channel operation.

3.4.1.7 Signal-to-Noise Plus Cross-Talk Ratio (see 4.8.23). The Signal-to-Noise Plus Cross-Talk Radio at each channel output terminal (near end or far end) due to full modulation on any other channel, shall exceed 53.0 db FIA weighting with two each multiplex equipments operated in a system. This value shall be achieved with the random noise source applied to all channels. For any one equipment aligned for normal operation, this ratio of 53.0 db shall be simultaneously met in at least ten of the channels in 12 channel operation and four of the channels in 6 channel operation, and all channels shall exceed 47.5 db. Those channels which exceed 55.0 db signal-to-noise ratio per 3.4.1.6 above, shall correspond to the channels that measure 53.0 db, and channels that exceed 49.0 db signal-to-noise shall correspond to the channels that measure 47.5 db.

3.4.1.8 Signal-to-Total Distortion (see 4.8.12). With the multiplexer aligned for 4 wire, 12 channel operation (see 3.4.1.3), the total distortion of a sine wave signal at full modulation and at any frequency between 300 and 3500 Hz shall be not less than 30.0 db below the fundamental. This shall apply for channels 1 through 11 and the distortion for channel 12 shall be not less than 24.0 db below the fundamental. (For six channel operation, 1 through 5 not less than 30.0 db and channel 6 not less than 24.0 db). In addition, the total distortion of a 1100 Hz sine wave at 23.0 db below full modulation shall not be less than 12.0 db below the fundamental for channels 1 through 11.

3.4.1.9 Delay Distortion (see 4.8.24). The delay distortion shall not exceed 200 microseconds over the band 600 to 3200 Hz. Delay distortion is measured as the difference between the maximum and minimum slopes of the phase vs. frequency characteristics. The delay measured at 1100 Hz is the reference delay.

3.4.1.10 Audio Impedance (see 4.8.25). The input and output impedance of each channel shall be 600 ohms ± 10% from 300 Hz to 3500Hz.

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3.4.1.11 Balance-to-Ground (see 4.8.26). The input and output terminals shall have not less than a 40.0 db balance-to-ground from 300 Hz to 3500 Hz.

3.4.2 Compression Ratio (see 4.8.32). The compression ratio, defined (see Figure 34) as the ratio of the initial to the average slope, shall be 20.0 db ± 1.0 db. In order to determine compliance with the compression ratio requirement, it is necessary that the compressor exhibit the characteristics as shown in Figure 34.

3.4.3 Coder-Decoder.

3.4.3.1 Coder (see 4.6). The coder shall be fully modulated by the compressor output. A six digit binary PCM code shall be employed. The nominal code level corresponding to zero modulation of each channel is midway between code levels 31 and 32. All code levels shall be between 29 and 34 for the test as specified in 4.8.8.

3.4.3.1.1 Code Level Stability (see 4.8.33). Channel code variation shall be no greater than two levels per 24 hour period from an initial code level between levels 29 (011101) and 34 (100010). No channel level shall vary from any other channel level by more than five code levels at any time. This test shall be performed at standard ambient conditions per MIL-STD-810.

3.4.3.1.2 Hum Level (see 4.8.27). The hum level at the input to the coder shall be less than 0.025% of the peak to peak PAM signal corresponding to a -4 dbm 1.1 KHz signal at the audio input.

3.4.3.2 Random Noise Source (see 4.8.34). The noise generator shall produce a noise output level of 4.5V ± 0.4 P-P. The spectrum of the generator output shall have a flatness of ± 4 db between 20 kHz and 200 kHz. The noise generator output level shall be 0.0 - 0.3V P-P when shorted. The noise generator output shall be monitored by the service facility, yellow band at specified output, hairline with generator shorted.

3.4.3.3 Squelch (see 4.8.13). While search and framing operations are taking place, the audio channels shall be quiet except that digital clicks may be present. The squelch shall operate upon initiation of the search operation.

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3.4.4 Frame Alarm (see 4.8.14). When the unit is not in frame, an audible alarm and a red light marked FRAME ALARM shall operate. The frame alarm shall also activate if timing to framing circuitry fails. A time delay of 0.5 to 5 seconds is incorporated into this alarm. A pushbutton switch shall turn off the audible alarm. When the equipment goes back into frame synchronization after the audible alarm has been turned off, the audible alarm will again operate until the audible alarm switch is returned to the initial position. Operation of the audible alarm on-off switch shall turn off the audible alarm without delay.

3.4.4.1 Master-Slave Switch (see 4.8.15). The master-slave switch shall operate one multiplexer as a master and a second multiplexer as a slave for 24 channel operation. The master unit shall operate on the (4 kHz) address and the slave unit shall operate on the slave (2 kHz) address for 24 channel operation. These units are interchangeable in master-slave operation by operation of the Master-Slave switch.

3.4.4.2 Separation (see 4.8.17). In 24 channel operation, Multiplexer TD-660( )/G shall separate the two interleaved 12 channel PCM signals. The signal containing the address selected by the MASTER-SLAVE switch shall be demodulated and the other signal shall be applied to the ALT PCM OUT connector.

3.4.5 Timing.

3.4.5.1 Transmit Timing (see 4.7.1, 4.8.8, and 4.8.1). Transmit timing shall be provided by an internal crystal controlled master oscillator operating at 576 kHz + 0.001 percent at room temperature with the TD-660 operating and stabilized. When operated over the temperature range specified in 4.6.1, the internal crystal controlled oscillator shall operate at 576 kHz + 0.003 percent. When operated at 104°F during the Electrical Preconditioning specified in 4.7.1, the oscillator shall operate at 576 kHz + 0.003 percent. In the slave mode of operation, the unit under test shall sync in with an external timing signal of 576 kHz ± .005 percent applied through the SYNC-IN connector.

3.4.5.2 Timing signal characteristics (see 4.8.8, 4.8.16 and 4.8.18). Input timing signal shall be those produced by Multiplexers TD-202( )/U, TD-204( )/U, TD-352( )/U, TD-660( )/G, TD-754( )/G and Radio Set AN/GRC-103( )(V) and may be ac or dc coupled. Output signals at the 12 CH TIM OUT and 6 CH TIM OUT connectors shall have a pulse peak 1.8 to 2.6 volts above a baseline of -2.0 to -2.3 volts. Output signals at the SYNC OUT XMTR AND SYNC OUT RCVR connectors shall have a pulse peak 1.6 to 2.4 volts above a baseline of -1.8 to -2.2 volts. Ripple at the baseline shall be within + 0.5 volts. The rise and fall time shall be 20 to 120 nanoseconds (ns) when measured from the 10% to the 90% amplitude

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points on the timing signal and the pulse width shall be 80 to 150 ns when measured at the 50% amplitude points on the leading and lagging edges of the timing signal. 12 CH TIM OUT signals shall lag SYNC IN signals by 0 to 100 ns. 12 CH TIM OUT shall lag 6 CH TIM OUT by 850 + 100 ns and shall coincide within + 40 ns with SYNC OUT XMTR. SYNC OUT RCVR shall lead ALT PCM OUT by 210 to 500 ns at 70° + 20°F and by 160 to 600 ns at other temperatures specified herein as operating temperature extremes. The 12 CH TIM OUT signal shall be at a 576 kHz rate for either 6 or 12 channel operation. The 6 CH TIM OUT signal shall be at 288 kHz rate for 6 channel operation. The 50% amplitude point on the leading edge of the timing signal shall lead the 50% point of the PCM transitions by the time ranges at the various temperatures as shown in the following phasing chart:

PHASING OF TIMING AND PCM SIGNALS

Temperature Range	Mode Number of Channels	Time Ranges (NS)
70° + 20°F and High Temperatures (+125°F)	6 and 12	-20 to +120
Low Temperature (-25°F)	6 12	-80 to +120 0 to +100

3.4.6 Pulse code modulation (PCM) signals.

3.4.6.1 PCM inputs and outputs. The PCM inputs and outputs shall be full width binary pulse trains.

3.4.6.2 PCM Signal characteristics (see 4.8.8 and 4.8.18). The 6/12 channel output of the transmitter shall be a full width binary pulse train. The pulse width of all PCM signals, when measured at the 50% amplitude points on the leading and lagging edges of the PCM signals, shall be 3.472 usec for 6 channel signals, 1.736 usec for 12 channel signals and 0.868 usec for 24 channel signals. The rise and fall times shall be 20 to 120 ns when measured from the 10% to the 90% amplitude points on the PCM signal. At the PCM OUT connector, digital "1" shall be a level between +0.0 and -0.4 volts and digital "0" shall be a level between -2.2 and -2.6 volts. Ripple at either level shall be less than + 0.4 volts peak. At ALT PCM OUT connector digital "1" level shall be between 0 and -0.4 V and digital "0" level shall be between -1.80 and -2.20 volts.

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3.4.7 Metering and controls (see 4.8.35). The service facility meter readings shall indicate within the hairline, green band, and yellow band as indicated by each color coded service facility switch position. The hairline adjustment indications shall be accurate to within +3%. The green and yellow tolerance bands shall be used as Go-No-Go indicators.

3.4.8 Digital impedance requirements (see 4.8.28). The input impedance of all those lines that receive signals and the load impedance of all those lines that transmit signals shall be 91 ohms ± 10%.

3.4.9 Channel monitoring (see 4.8.5, 4.8.21, and 4.8.29). The operator shall talk over and monitor each audio channel through the use of Handset H-156( )/U. The output of the talk circuit shall fully modulate the channel. The output of the Earphone Monitor with a +1 dbm signal applied to the channel output shall be -10.0 ± 3.0 dbm into 150 ohms (NOTE: Add +6 db to readings taken with meter calibrated in dbm into 600 ohms). The talk-monitor facilities shall operate without interfering with normal traffic during any section four test. The response of the microphone amplifier and the earphone amplifier shall each be ±2.0 db from 300 to 3500 Hz, when tested in accordance with 4.8.29.

3.4.10 Channel Alignment (see 4.8.19 and 4.8.30). Alignment of audio channels shall be by use of the integrated test facilities. These shall include an internal sinewave oscillator and the test align meter. The oscillator frequency shall be 1100 ± 90 Hz, output level shall be equivalent to a channel input level of -4.0 ± 0.5 dbm, and total distortion shall be less than 26db below output level. The test align meter and circuitry shall permit adjustment of channel output levels to -4.0 ± 0.5 dbm for 4 wire operation and +1.0 ± 0.5 dbm for 2 wire operation with full modulation input. The above tolerances include operation over the temperature range specified in 4.6.1. The 2 WIRE / 4 WIRE switch shall be used to change the meter hairline calibration for either the 2 WIRE or 4 WIRE output levels listed above. The overall alignment tolerance using the internal test facilities shall be ± 1.0 db. During talk and listen inspection all channels tested shall be undistorted without evidence of digital clicks.

3.4.11 Power Supply 11A1/11A33.

3.4.11.1 AC Input (see 4.8.36). The equipment shall meet the requirements of 3.4.1.6, 3.4.1.8, and 3.4.3.1.1 for the tests specified in 4.8.36 when energized from 115 volts ac + 5%, 50 to 400 Hz + 5%. The power input of the Multiplexer TD-660( )/G shall not exceed 80 watts using the 11A1 power supply or 55 watts for the 11A33 power supply.

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3.4.11.2 DC Output (see 4.8.37). The internal power supply shall provide the following listed dc outputs over the temperature range specified in 4.6.1. Ripple measurements shall be performed at the power supply front panel test points.

11A1/11A33 Voltage (volts)	Tolerance (percent)	Ripple (Maximum) (Millivolts rms)
+26.7	+20.0	--
+12.00	±3.0	23
+7.00	±0.5	15
+4.10	±7.0	20
-4.40	±5.0	23
-6.20	±5.0	23
-12.10	±5.0	23

3.4.12. AUXILIARY Signal Characteristics (see 4.8.20). Four output signals and one input signal shall appear at the AUX connectors located on the rear bulkhead of the unit. These signals and their characteristics are described in 3.4.12.1 through 3.4.12.5 inclusive. The pulse width shall be measured at the 50% amplitude points, and the rise and fall times shall be measured from the 10% amplitude to the 90% amplitude points on each auxiliary signal.

3.4.12.1 AUX ADDRESS (see 4.8.20.h). This output signal is a negative going pulse that repeats at an 8 kHz rate. The pulse width shall be 1.74 usec for 12 channel operation and 3.48 usec for 6 channel operation. The base line of this signal shall be +0.0 to -0.4 volts and the negative peak shall be -2.2 to -2.6 volts. Rise and fall times shall be 20 to 120 ns. This signal shall be coincident with the address digit in the PCM OUT train to within ± 40 ns. Ripple at either level shall be within ±0.5 volts.

3.4.12.2 RCVR AUX CLOCK (see 4.8.20.h). This output signal is a series of positive going pulses that occur at a 576 kHz rate in 12 channel operation and at a 288 kHz rate in 6 channel operation. The pulse width shall be 80 to 150 ns. The base line of this signal shall be -2.2 to -2.6 volts and the pulse peak shall be +0.0 to -0.4 v. Rise and fall time shall be 20 to 120 ns. These pulses shall be disabled when the multiplexer receiver has lost frame synchronization. Out of frame voltage level is -2.2 to -2.6 v. This signal shall lead the PCM to AUX signal by 0 to 80 ns. Ripple at base line shall be within ± 0.5 volts.

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3.4.12.3 RCVR AUX ADDRESS (see 4.8.20.h). This output signal is a positive going pulse that repeats at an 8 kHz rate. The pulse width shall be 80 to 150 ns. The baseline shall be -2.2 to -2.6 volts, and the pulse peak shall be +0.0 to -0.4 volts. The rise and fall time shall be 20 to 120 ns. The leading edge of this pulse shall lead the leading edge of the frame pulse in the RCVR AUX CLOCK signal by 0 to 160 ns. Ripple at baseline shall be within ±0.5 volts.

3.4.12.4 PCM TO AUX (see 4.8.20.h). This output signal is the framed PCM signal sent to the AUX unit. It occurs at a 576 kHz bit rate in 12 channel operation and at a 288 kHz bit rate in 6 channel operation. The signal amplitude shall be +0.0 to -0.4 volts for a logical "1" and -2.2 to -2.6 volts for a logical "0". Pulse width shall be 1.736 usec for 12 channel operation and 3.472 usec for 6 channel operation. Rise and fall times shall be no greater than 120 ns. This signal shall lead the negative transition of the decoder serial clock (pin 30 on panel 11A28) by a maximum of 250 ns at the 50% amplitude points. The ripple at either level shall be within ±0.5 volts.

3.4.12.5 PCM FROM AUX (see 4.8.20.t). This input signal is an incoming PCM signal from the AUX unit. This signal will have the same characteristics as the PCM to AUX signal (see 3.4.12.4) except phasing and signal level. This signal shall be delayed by 300 to 980 ns with respect to the PCM TO AUX signal. Digital "1" level shall be 0.0 to -0.4 volts and digital "0" shall be -1.6 to -3.0 volts. The multiplexer shall perform satisfactorily when receiving the incoming PCM signal, with variations in characteristics as described above, from the AUX unit.

3.5 Preconditioning.

3.5.1 Electrical Preconditioning (Burn-In)(see 4.5 and 4.7.1). Prior to submission to the specified quality conformance inspections, each assembled Multiplexer TD-660( )/G consisting of the chassis, assembly, all panel assemblies and power supply, plus spare assemblies defined by contract, shall be electrically preconditioned for a minimum of 48 hours to a maximum of 168 hours operating time. The last 24 hours of electrical preconditioning shall be failure free. Each fabricated TD-660( )/G and spare assemblies shall be submitted to this process.

3.5.2 Bounce Preconditioning (see 4.5 and 4.7.2). Each TD-660( )/G assembly shall meet the requirements herein, without subsequent processing after submission to the bounce preconditioning of 4.7.2. Each TD-660( )/G fabricated shall be subjected to this process.

3.6 Service Conditions (see 4.6). The equipment shall meet the Service Conditions listed below.

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3.6.1 Temperature. The following low and high temperature requirements shall be met, during and at the conclusion of the temperature tests. There shall be no evidence of electrical or mechanical degradation.

3.6.1.1 Low. During testing as specified in 4.6.1.1, the equipment shall meet the performance specified in 4.8.8. During and after testing, the equipment shall meet full specification performance for the measurements as specified in 4.8.8.

3.6.1.2 High. During testing as specified in 4.6.1.2, the equipment shall meet the performance specified in 4.8.8. During and after testing, the equipment shall meet full specification performance for the measurements as specified in 4.8.8.

3.6.2 Humidity (Moisture Resistance)(see 4.6.2). During testing as specified in 4.6.2, the equipment shall meet the performance requirements of 4.8.7. After testing, the equipment shall meet full specification performance for the measurements as specified in 4.8.7. At the conclusion of this test, the equipment shall not exhibit rust and/or deterioration of parts and materials greater than permitted by the individual component drawings, specifications, and subsidiary equipment finish specifications. The dip brazed chassis joints shall not exhibit white salt deposits in excess of one trace of salt deposits per linear inch, not to exceed 50% of the brazed fillet width, exceed 3/16 inch in length nor cover a total area of 0.030 inch.

3.6.3 Altitude (Elevation)(see 4.6.3). The equipment shall meet full specification performance of 4.8.7 during and after testing as specified in 4.6.3.

3.6.4 Vibration, Bounce, and Shock.

3.6.4.1 Vibration, Resonance Search (see 4.6.4). The equipment shall meet the performance requirements of 4.8.6 when tested in accordance with 4.6.4, frequency range 10 - 55 Hz. There shall be no evidence of electrical or mechanical degradation.

3.6.4.2 Shock, Bounce, Vehicular (see 4.6.5). The equipment shall meet requirements of 4.8.6 after being tested in accordance with 4.6.5. There shall be no evidence of electrical or mechanical degradation.

3.6.4.3 Shock (Bench Handling)(see 4.6.6). The equipment in its servicing condition, except the Card Cell Cover shall be installed, shall withstand shock encountered during servicing and meet the requirements of 4.8.6 when tested in accordance with 4.6.6. There shall be no evidence of electrical or mechanical degradation.

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3.6.5 Fungus (see 4.6.10). After the fungus test of 4.6.10, the equipment shall be visually examined, using a 10-power magnifier. The equipment shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.3) as evidenced by growth colonization (see 6.3)(which includes branching (see 6.3) and sporulation (see 6.3)) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.3) shall not be included in this evaluation.

3.6.6 Dripproof (see 4.6.7). The equipment in normal operating condition shall meet the requirements of 4.8.7 when tested in accordance with 4.6.7. The equipment shall not leak or show evidence of leakage or electrical degradation at the conclusion of the test.

3.6.7 Salt Fog (see 4.6.8). After the salt fog test of 4.6.8, the equipment, when examined visually with the aid of a 10-power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering, or loosening of finish or metal surface; or exfoliation (see 6.3) of metal. Electrical operation is not required.

3.6.8 Dust (see 4.6.9). After testing as specified in 4.6.9, the equipment shall meet the tests of 4.8.6. Except for exterior paint damage on the case or front panel, the equipment shall exhibit no electrical or mechanical damage. The equipment shall not show evidence of dust entry into the interior of the TD-660( )/G.

3.7 TEMPEST (see 4.13). The TD-660( )/G shall meet the TEMPEST requirements of MIL-T-55654.

3.8 Finish, Protective. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings (see 4.4).

3.9 Marking.

3.9.1 General. Marking shall conform to Specification MIL-M-13231. (See 4.4). Front panel marking shall be Group I as described in that specification.

3.9.2 Visibility. Wherever practicable, parts shall be mounted so that their identification markings will be readily visible with minimum disassembly of the equipment.

3.9.3 Serial Numbers (see 4.4 Marking). Each Multiplexer TD-660( )/G shall be serial numbered.

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3.10 Microelectronic Integrated Circuits (see 4.4). All microelectronic integrated circuit devices used in this equipment shall meet the requirements of MIL-M-38510 for Class B devices and their individual drawing requirements.

3.11 Electromagnetic Interference Characteristics (see 4.10). The equipment shall meet the electromagnetic interference requirements specified in MIL-STD-461, for Class 1C equipment. Tests and test methods shall be as specified in MIL-STD-462.

3.12 Interchangeability (see 4.8.30 and 4.11). Like units, assemblies, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.13 Panel Assemblies (Electrical Test)(see 4.9). The TD-660( )/G and panel assemblies listed below shall meet the requirements of their respective test requirement drawings as specified in 4.9.

Multiplexer TD-660( )/G (TRS Test and Alignment Requirements) SM-B-525082

Panel Assembly 11A13	SM-B-524691	Panel Assembly 11A28	SM-B-525262
Talk Monitor 11A21	SM-B-525110	Panel Assembly 11A29	SM-B-525267
Filter Assembly 11A18	SM-B-525112	Filter Assembly 11A31	SM-B-525218
Panel Assembly 11A18A1	SM-B-525115	Panel Assembly 11A31A1	SM-B-525214
Filter Assembly 11A19	SM-B-525090	Panel Assembly 11A31A2	SM-B-525219
Panel Assembly 11A23	SM-B-525237	Filter Assembly 11A32	SM-B-525224
Panel Assembly 11A24	SM-B-525242	Panel Assembly 11A32A1	SM-B-525225
Panel Assembly 11A25	SM-B-525247	Panel Assembly 11A32A2	SM-B-525230
Panel Assembly 11A26	SM-B-525252	Power Supply 11A33	SM-B-317431
Panel Assembly 11A27	SM-B-525257	Panel Assembly 11A33A1	SM-B-317432

3.14 Safety Factors (see 4.12).

3.14.1 Radioactive materials shall not be employed in the TD-660( )/G.

3.14.2 Finished items shall not contain any raw, sharp, or rough edges on any parts.

3.15 Operational Test (see 4.8.5). Each fabricated and tested Multiplexer TD-660( )/G shall be subjected to a Final Operational Test in accordance with 4.8.5 after all other inspections and tests are completed to determine that the TD-660( )/G is operational prior to packaging. The requirements of 3.4.1.6, 3.4.1.8, 3.4.5.1, 3.4.7, 3.4.11.2, and 3.4.9 shall be met. (NOTE: Two each TD-660( )/G's shall be connected for Electrical Preconditioning Test, paragraph 3.5.1, Figure 1.)

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3.16 Workmanship (see 4.4). The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraph:

In MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
  Slack.
  Protection.
  Clearance.
  Splicing.
  Connections, general.
  Grounding, general.
  Shielding on wire and cable.
Soldering and brazing.
Cleaning.
General requirements for securing of parts.
Cleaning prior to welding.
Process for welding.
Cleaning after welding.
Controls.
Self-tapping screws.

3.17 Printed wiring assemblies (see 4.4). Printed wiring assemblies shall be in accordance with the applicable paragraph in MIL-P-11268 and shall be conformally coated in accordance with MIL-STD-275 with a coating material that conforms to MIL-I-46058, type ER (Epoxy).

3.18 Reliability (see 4.14). Each equipment on order shall have a specified mean-time-between-failure (θo) of 1190 hours. The testing as specified in 4.14 herein, will be conducted to determine compliance to the reliability requirement above.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Test equipment and facilities for Government Inspection shall be made available by the contractor.

4.2 Classification of Inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3) - does not include preparation for delivery.

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.15).

4.3 First Article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First Article Units. The contractor shall furnish 16 first article units of the complete Multiplexer TD-660( )/G.

4.3.2 First Article Inspection. The first article inspection shall consist of inspections specified in Table I, and shall be performed in the order specified herein.

4.3.3 First Article Data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

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TABLE I - FIRST ARTICLE INSPECTION

Inspections	Rqt Para	Test Para	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6	Unit 7	Unit 8	Units 9 - 16
1. Inspections covered by subsidiary documents		4.4	Inspection to be performed on all units								
2. Bounce preconditioning	3.5.2	4.7.2	Inspection to be performed on all units								
3. Electrical preconditioning	3.5.1	4.7.1	Inspection to be performed on all units								
4. Group A Inspection	(See Table II)		Inspection to be performed on all units								
5. Group B Inspection	(See Table III)		Inspection to be performed on all units								
6. TEMPEST	3.7	4.13	1								
7. EMI	3.11	4.10					1				
8. Group C Inspection											
Low Temperature	3.6.1.1	4.6.1.1		1					1		
High Temperature	3.6.1.2	4.6.1.2		2					2		
Humidity	3.6.2	4.6.2						2			
Altitude (elevation)	3.6.3	4.6.3			1						
Vibration	3.6.4.1	4.6.4			2						
Bounce (vehicular)	3.6.4.2	4.6.5			3					2	
Shock (bench handling)	3.6.4.3	4.6.6			4						
Fungus (See Note D)	3.6.5	4.6.10				1					
Dripproof	3.6.6	4.6.7						1		1	
Salt Fog (See Note D)	3.6.7	4.6.8		4							
Dust	3.6.8	4.6.9		3							
9. Group D Reliability	3.18	4.14									1 (Note C)

NOTES:

a. The inspections 1 through 5, in the order shown, shall be performed on all First Article units before subjecting these units to any other inspection requirements in the table.

b. The numbers in the unit columns in the table covered by Inspections numbered 6, 7, and 8, specify the order of inspections for the indicated unit.

c. Units 9 - 16: These units are to undergo Inspections 1, 2, 3, 4, 5, and 9.

d. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

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4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement in the contract or purchase order.

Item	Where Required
Parts, Materials, and Processes; General	3.3
Finish	3.8
Marking	3.9
Microelectronic Integrated Circuits	3.10
Workmanship	3.16
Printed Wiring Assemblies	3.17

4.5 Quality Conformance Inspection of Equipment Before Preparation For Delivery. The contractor shall perform the inspections specified in 4.4, 4.7, and 4.5.1 through 4.5.5 in that order. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A Inspection. Each unit (and each subassembly identified in 3.13) on contract or purchase order shall be inspected for conformance to the inspections specified in Table II (Manufacturing Inspection). Units shall be comprised of subassemblies which have been subjected to and passed the Manufacturing Inspection and Quality Assurance Verification cited herein. Discrete lost for Quality Assurance Verification shall be formed from units and subassemblies that pass the Manufacturing Inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be submitted to sampling inspection, utilizing the procedures of MIL-STD-105 using general inspection levels and AQL's indicated in Table II (Quality Assurance Verification). Group A inspection shall be performed in any order which is satisfactory to the Government, except that the Operational Test, 4.8.5, shall be last (see 6.4).

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TABLE II - GROUP A INSPECTION

Inspection	Req Para	Ref Fig	Insp Para	AQL MAJOR	AQL MINOR
Visual-Mechanical					
TD-660( )/G Case & Covers	3.16		4.12	2.5*	10.0*
Filter Assembly 11A18/11A18A1	3.16		4.12	2.5*	4.0*
Filter Assembly 11A19	3.16		4.12	2.5*	4.0*
Panel Assembly 11A13	3.16		4.12	2.5*	4.0*
Talk Monitor 11A21	3.16		4.12	2.5*	4.0*
Panel Assy 11A23	3.16		4.12	1.0*	4.0*
Panel Assy 11A24	3.16		4.12	1.0*	4.0*
Panel Assy 11A25	3.16		4.12	1.0*	4.0*
Panel Assy 11A26	3.16		4.12	1.0*	4.0*
Panel Assy 11A27	3.16		4.12	1.0*	4.0*
Panel Assy 11A28	3.16		4.12	1.0*	4.0*
Panel Assy 11A29	3.16		4.12	1.0*	4.0*
Filter Assy 11A31	3.16		4.12	1.0*	4.0*
Panel Assy 11A31A1	3.16		4.12	1.0*	4.0*
Panel Assy 11A31A2	3.16		4.12	1.0*	4.0*
Filter Assy 11A32	3.16		4.12	1.0*	4.0*
Panel Assy 11A32A1	3.16		4.12	1.0*	4.0*
Panel Assy 11A32A2	3.16		4.12	1.0*	4.0*
Power Supply 11A33	3.16		4.12	1.0*	4.0*
Panel Assy 11A33A1	3.16		4.12	1.0*	4.0*
*Defects per Hundred Units					
Electrical Performance (Including spare panel assys)					
Panel Assy 11A23	3.13		4.9	1.0%	**
Panel Assy 11A24	3.13		4.9	1.0%	**
Panel Assy 11A25	3.13		4.9	1.0%	**
Panel Assy 11A26	3.13		4.9	1.0%	**
Panel Assy 11A27	3.13		4.9	1.0%	**
Panel Assy 11A28	3.13		4.9	1.0%	**
Panel Assy 11A29	3.13		4.9	1.0%	**
Panel Assy 11A13	3.13		4.9	1.0%	**
Panel Assy 11A31A1	3.13		4.9	1.0%	**
Panel Assy 11A31A2	3.13		4.9	1.0%	**
Filter Assy 11A31	3.13		4.9	1.0%	**
Panel Assy 11A32A1	3.13		4.9	1.0%	**
Panel Assy 11A32A2	3.13		4.9	1.0%	**
**All electrical defects are major.					

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TABLE II - GROUP A INSPECTION (CONTD)

Inspection	Req Para	Ref Fig	Insp Para	AQL MAJOR	AQL MINOR
Electrical Performance (contd)					
Filter Assy 11A32	3.13		4.9	1.0%	**
Panel Assy 11A18A1	3.13		4.9	1.0%	**
Filter Assy 11A18	3.13		4.9	1.0%	**
Filter Assy 11A19	3.13		4.9	1.0%	**
Panel Assy 11A33A1	3.13		4.9	1.0%	**
Power Supply 11A33	3.13		4.9	1.0%	**
Talk Monitor 11A21	3.13		4.9	1.0%	**
Multiplexer TD-660( )/G	3.13		4.9	1.0%	**
(Test and alignment drawing requirements)					
Electrical					
Signal-to-Noise Ratio	3.4.1.6	2	4.8.9	1.0%	**
Frequency Response	3.4.1.1	4	4.8.10	1.0%	**
Output Level	3.4.1.3	5	4.8.11	1.0%	**
Signal-to-Total Distortion	3.4.1.8	3	4.8.12	1.0%	**
Digital Squelch	3.4.3.3	6	4.8.13	1.0%	**
Frame Alarm	3.4.4	7	4.8.14	1.0%	**
Master-Slave Operation	3.4.4.1	8	4.8.15	1.0%	**
Sync-In	3.4.5.2	9	4.8.16	1.0%	**
Separation	3.4.4.2	10	4.8.17	1.0%	**
Digital Signal	3.4.6.2	11	4.8.18	1.0%	**
Channel Alignment	3.4.10	13, 14	4.8.19	1.0%	**
AUX ADDRESS	3.4.12.1	12	4.8.20h	1.0%	**
RCVR AUX CLOCK	3.4.12.2	12	4.8.20h	1.0%	**
RCVR AUX ADDRESS	3.4.12.3	12	4.8.20h	1.0%	**
PCM TO AUX	3.4.12.4	12	4.8.20h	1.0%	**
PCM FROM AUX	3.4.12.5	15	4.8.20h	1.0%	**
Operational Test	3.15	16	4.8.5	1.0%	**
Signal-to-Noise Ratio	3.4.1.6	16	4.8.5	1.0%	**
Signal-to-Total Distortion	3.4.1.8	16	4.8.5	1.0%	**
Transmit Timing	3.4.5.1	16	4.8.5	1.0%	**
Metering	3.4.7	16	4.8.5	1.0%	**
Channel Monitoring	3.4.9	16	4.8.5	1.0%	**
DC Output	3.4.11.2	16	4.8.5	1.0%	**
**All electrical defects are major.					

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4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in the order listed in Table II.

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

TABLE III - GROUP B INSPECTION

Inspection	Req Para	Ref Fig	Insp Para	AQL
Interchangeability				
Mechanical	3.12 .		4.11	6.5%
Electrical	3.12	16	4.8.30	6.5%
Input Level	3.4.1.2	5	4.8.21	4.0%
Input-Output Linearity	3.4.1.4	5	4.8.22	4.0%
Audio Level Stability	3.4.1.5.1	4	4.8.31	6.5%
Signal-to-Noise plus Crosstalk Ratio	3.4.1.7	17	4.8.23	4.0%
Delay Distortion	3.4.1.9	18	4.8.24	4.0%
Audio Impedance	3.4.1.10	19	4.8.25	4.0%
Balance to Ground	3.4.1.11	20	4.8.26	4.0%
Compresser Characteristics	3.4.2	21, 34	4.8.32	6.5%
Code Level Stability	3.4.3.1.1	22	4.8.33	6.5%
Hum Level	3.4.3.1.2	23	4.8.27	4.0%
Random Noise Source	3.4.3.2	25	4.8.34	6.5%
Timing Signal Char.	3.4.5.2	9	4.8.16	4.0%
PCM Signal Char.	3.4.6.2	11	4.8.18	4.0%
Metering	3.4.7	27	4.8.35	6.5%
	3.4.10	28	4.8.35	6.5%
Digital Impedance	3.4.8	26	4.8.28	4.0%
Channel Monitoring	3.4.9	29,30	4.8.29	4.0%
AC Power Supply Input	3.4.11.1	31	4.8.36	4.0%
DC Power Supply Output	3.4.11.2	31	4.8.37	4.0%

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4.5.2.1 Group B Sampling Plans. The Group B AQL shall be as shown in Table III and the inspection level shall be S-4.

4.5.2.2 Order of Inspection Within Group B. Group B inspection shall be performed in the order listed in Table III.

4.5.3 Group C Inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

TABLE IV - GROUP C INSPECTION

Inspection	Req Para	Ref Fig	Insp Para
Sub-Group I			
Temperature			
Audio Level Stability at Temperature Extremes	3.4.1.5.2	32	4.6.1, 4.8.8
Timing and PCM Signal Characteristics at Temperature Extremes	3.4.5.2, 3.4.6.2	32	4.6.1, 4.8.8
Coder	3.4.3.1	32	4.8.8
Low	3.6.1.1	32	4.6.1.1, 4.8.8
High	3.6.1.2	32	4.6.1.2, 4.8.8
Humidity	3.6.2	32	4.6.2, 4.8.7
Sub-Group II			
TEMPEST	3.7		4.13
Sub-Group III			
Dripproof	3.6.6	33	4.6.7, 4.8.7
Vibration	3.6.4.1	33	4.6.4, 4.8.6
Bounce, Vehicular	3.6.4.2	33	4.6.5, 4.8.6
Shock, Bench Handling	3.6.4.3	33	4.6.6, 4.8.6
EMI	3.11	24	4.10
Sub-Group IV			
Altitude (elevation)	3.6.3	32	4.6.3, 4.8.7
Dust	3.6.8	33	4.6.9, 4.8.6 (Note 1)
Fungus	3.6.5	33	4.6.10, 4.8.6 (Note 1)
Salt Fog	3.6.7	33	4.6.8 (Note 1)

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried and refurbished if necessary, before proceding with subsequent tests.

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4.5.3.1 Sampling for Group C inspection.

4.5.3.1.1 Sub-group I. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every month. Each unit shall be subjected to temperature and humidity testing in that order.

4.5.3.1.2 Sub-group II. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every month. Each unit shall be subjected to TEMPEST testing.

4.5.3.1.3 Sub-group III. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every third month. One unit shall be subjected to EMI, Dripproof and Shock, Bench Handling testing, and one unit shall be subjected to Vibration (Resonance Search), and Bounce Vehicular testing with each unit subjected to testing in the order given.

4.5.3.1.4 Sub-group IV. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every fourth month. One unit shall be subjected to Altitude (Elevation), Dust and Salt Fog testing, in that order. One unit shall be subjected to fungus testing.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in the order specified in 4.5.3.1.

4.5.3.3 Non-compliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of the failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative (QAR). After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the correction action was effective.

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4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.5.5 Group D inspection. This inspection shall consist of the tests specified in Table V and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection. Test conditions, methods, and procedures shall be in accordance with paragraph 4.14.

TABLE V - GROUP D INSPECTION

Inspection	Req Para	Insp Para
Reliability	3.18	4.14

4.5.5.1 Sampling for Group D inspection. Eight samples shall be selected at random from the first production lot. For subsequent Group D inspection, eight units shall be selected at random from each month's production lot.

4.5.5.2 Group D failures. A Group D failure is defined as any deviation in performance from the requirements specified in Table V.

4.5.5.3 Non-Compliance. In case of Group D failure occurrence, procedures and requirements specified elsewhere in contract shall be followed.

4.5.5.4 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Service Condition Tests. The following service condition tests shall be performed. Since more than one unit is available, some tests may be run concurrently with Government approval by the Contracting Officer or his designated representative. After completion of the Fungus and Salt Fog tests, the equipment must be thoroughly washed, cleaned, dried and refurbished, if necessary, before proceeding with subsequent tests.

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4.6.1 Temperature Tests.

4.6.1.1 Low Temperature. (See 3.6.1.1) The equipment shall be subjected to Test Method 502.1, Procedure I of MIL-STD-810. Storage temperature shall be -65 degrees F and operation temperature shall be -25 degrees F. Storage time shall be 24 hours. At operation temperature, tests shall start within 5 minutes after ac power application. A pretest performance record is required and is specified in 4.8.8. Performance checks per 4.8.8 shall be made as specified in Test Method 502.1 during tests.

4.6.1.2 High Temperature (see 3.6.1.2). The equipment shall be subjected to Test Method 501.1, Procedure II of MIL-STD-810 except as follows: Step 4, the non-operating temperature is 68°C (154°F), Step 8 shall be modified as follows: Solar radiation of the intensity of 360 BTU/sq. ft/hr shall be maintained for four hours and then tested in accordance with paragraph 4.8.8. The highest operating temperature is 125° +5° - 0°F not including the effects of solar radiation. Performance tests per 4.8.8 shall be made during tests as specified in Test Method 501.1, Procedure II.

4.6.2 Humidity (Moisture-Resistance) Test (see 3.6.2). The equipment with the power supply, top, rear, and panel covers removed shall be subjected to Test Method 507.1, Procedure II of MIL-STD-810. Test measurements per 4.8.7 shall be taken during the 24 to 28 hr period and the last 5 hr period during each 48 hour cycle. Insert the Power Supply in the end item for electrical tests. At conclusion of test, the equipment shall meet full specified performance when tested in accordance with 4.8.7.

4.6.2.1 Failure. If any equipment fails to meet the performance specified in 4.8.7 during cycling, it shall be realigned or readjusted once. If the equipment then fails or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.8.7, it does not pass the test.

4.6.3 Low pressure (Altitude)(see 3.6.3). The equipment shall be subjected to the following test:

a. The equipment shall be placed in the test chamber. Prior to testing, the equipment shall meet the requirements for the tests specified in 4.8.7 except that auxiliary test equipment need not be inclosed in the chamber. A temperature of 73° ± 18°F shall be maintained during the entire test.

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b. Reduce the chamber internal pressure to 87.5 mm of Hg (3.44 in of Hg or 50,000 feet above sea level). The rate of pressure change may be the maximum attainable by the chamber. Maintain the chamber pressure for a period of not less than one hour.

c. Increase the chamber pressure to 523 mm of Hg (20.6 in of Hg or 10,000 feet above sea level) and then perform the test of 4.8.7 and record results.

d. With the equipment not operating, return the chamber to standard ambient conditions and perform the test of 4.8.7 and visually inspect the equipment.

e. During and after the test, the equipment shall meet the requirements for the test of 4.8.7 and shall not exhibit any electrical or mechanical degradation which could in any manner prevent the equipment from meeting any requirements of this specification.

4.6.4 Vibration Test (Resonance Search)(see 3.6.4.1).

a. Resonant frequencies of the equipment shall be determined by varying the frequency of applied vibration slowly through the specified range at reduced test levels but with sufficient amplitude to excite the item.

b. The test item shall be attached to the vibration exciter table by its normal mounting means or by means of a rigid fixture capable of transmitting the vibration conditions. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of undesirable responses in the test.

c. The test item shall be operated during application of vibration resonance search so that functional effects caused by these tests may be evaluated.

d. The test level shall be applied over a frequency range of 10 to 55 cps and at a vibratory displacement of not less than 0.030 inches, double amplitude. The vibratory frequency shall be changed in discrete steps of 1 cps and maintained at least 10 seconds at each frequency. Where possible, fixturing shall be such that resonances within the test item can be observed and measured. The test item shall have no resonances in the frequency range of 10 to 55 cps that exceed twice the amplitude of applied vibration.

e. The equipment shall meet pretest and post test performance requirements in accordance with paragraph 4.8.6.

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4.6.5 Bounce (Vehicular) Test (see 3.6.4.2). The equipment shall be subjected to Test Method 514.2, Procedure IX, Part 2 of MIL-STD-810. At the conclusion of test, the equipment shall meet full specified performance when tested in accordance with 4.8.6. Pretest data per 4.8.6 is required.

4.6.6 Shock (Bench Handling) Test (see 3.6.4.3). The equipment shall be subjected to Test Method 516.2, Procedure V of MIL-STD-810. At the conclusion of test, the equipment shall meet full specified performance when tested in accordance with 4.8.6. Pretest data per 4.8.6 is required. The Card Cell Cover shall be installed during all drops. The equipment shall not be dropped on its front face.

4.6.7 Dripproof Test (see 3.6.6). The equipment shall be tested per MIL-STD-108, 4.3(b) Drip Test, except as follows:

a. In lieu of equipment orientation 15° from the vertical, drops of liquid shall be falling at an angle of 15° from the vertical when the equipment is mounted in its normal position.

b. While the water is being applied, the equipment shall be connected per Figure 33. During each operating period, the audio gain on one channel with -4 dbm input shall be recorded once each minute. The tests of 4.8.7 shall be met.

c. Immediately following the drip test, the equipment shall meet the full specified performance when tested in accordance with 4.8.7.

4.6.8 Salt Fog Test (see 3.6.7). The equipment shall be subjected to the salt fog test specified in Method 509.1, Procedure I, of MIL-STD-810. No corrosion products shall be deposited on the test item by the facility. Inability of the equipment to meet the requirements of 3.6.7 shall constitute failure of this test.

4.6.9 Dust Test (see 3.6.8). The equipment, in its operating configuration, shall be subjected to Test Method 510.1, Procedure I of MIL-STD-810. The equipment shall not be operated during the test. Pretest and post test data is required in accordance with 4.8.6.

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4.6.10 Fungus Test (see 3.6.5). The equipment shall be subjected to the fungus test specified in Method 508.1, Procedure I of MIL-STD-810. There shall be abundant growth colonization (6.3) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the equipment is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment. Inability of the equipment to meet the requirements of 3.6.5 shall constitute failure of this test.

4.7 Preconditioning (see 3.5).

4.7.1 Electrical Preconditioning (Burn-In) Inspection (see 3.5.1). Electrical preconditioning inspection shall meet the requirements of 3.5.1.

4.7.1.1 Test Details.

Electrical Preconditioning. The electrical preconditioning shall be conducted. Test level B of MIL-STD-781 shall apply for temperature and vibration requirements. Each panel, including spares, shall also meet the requirements of 3.5.1 Preconditioning.

a. During testing, the equipment shall be mounted on a vibration table, positioned in its normal operating plane with all covers in place. Vibration shall be in a plane perpendicular to the 11A23 and 11A29 printed circuit boards. During all operating periods, the equipment under test shall be electrically connected per Figure 1. The equipment shall be turned on and operated in the 12 channel mode for 23 hours and then turned off for 1 hour, the next 24 hour period, 23 hours operating and 1 hour off, shall be in the 6 channel mode as indicated in Figure 1. Time expended to perform the operational tests shall be considered part of the "ON" period of the "ON-OFF" cycle and shall not extend beyond the twenty-third hour. For these tests, 6 channel and 12 channel modes shall be used alternately. At the end of each operating period, the mode shall be changed. Prior to the start and at the end of each 24 hour period (6 or 12 channel mode), the operational test (6 or 12 channel mode) of 4.8.5 (Table VI) shall be performed and the test data recorded.

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TABLE VI

Test	Req Para	Fig	Test Para
Operational Test	3.15	1, 16	4.8.5
Signal-to-Noise Ratio	3.4.1.6	1, 16	4.8.5
Signal-to-Total Distortion	3.4.1.8	1, 16	4.8.5
Transmit Timing	3.4.5.1	1, 16	4.8.5
Metering	3.4.7	1, 16	4.8.5
Channel Monitoring	3.4.9	1, 16	4.8.5
DC OUTPUT	3.4.11.2	1, 16	4.8.5

NOTE: For the Operational Test conducted in Table VI, two each TD-660's shall be connected together for 6/12 channel operation in lieu of one each TD-660 connected in loop-back mode.

b. The chassis, plug-in printed circuit panel assemblies and power supplies shall be considered to have been successfully electrically preconditioned when they have individually passed a cumulative 48 hour operating preconditioning period without a failure during the last 24 hour period. If during the last 24 hour operating preconditioning period a plug-in panel assembly, power supply, or chassis fails, it may be replaced with a preconditioned panel, power supply, or chassis and the preconditioning continued without the restart of a new 48 hour preconditioning period. A plug-in panel assembly, power supply or chassis failing operating preconditioning shall, after repair, be resubmitted to and pass the entire 48 hour operation preconditioning period prior to final electrical assembly tests by the contractor.

c. The unit chassis shall have no less than 48 hours of operational preconditioning nor more than 168 hours of operational preconditioning (exclusive of production test time) when used as a vehicle for preconditioning power supplies and plug-in assemblies (including all spares).

d. Assemblies failing electrical acceptance tests subsequent to satisfactory completion of electrical preconditioning may be repaired with an electrically preconditioned assembly and electrically retested, accepted and delivered without repeat of electrical preconditioning.

4.7.1.2 Marking of electrically preconditioned units. All electrically preconditioned plug-in panel assemblies, power supplies, and chassis shall be physically identified by a preconditioning stamp. The stamp shall be affixed to the front panel near the nameplate of each unit which has successfully completed the preconditioning as outlined herein. The stamp on preconditioned plug-in panel assemblies shall be affixed to the bottom

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side, and to the bottom outside edge of the power supply subassembly 11A33. The stamp shall be a "p" in a circle and shall be marked on the preconditioned unit per the Group III requirements of MIL-M-13231.

4.7.2 Bounce Preconditioning (see 3.5.2). Each unit, in its non-operating condition, shall be placed in its normal operating position on the table of the package tester Type 1000 SC as made by the L. A. B. Corporation, Skaneatecles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. Each unit of each lot of equipment shall be subjected to this preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.8 Electrical Tests.

4.8.1 Standard Conditions. Unless otherwise specified, all tests shall be performed at 115 volts + 5 percent from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperatures, pressure and humidity per MIL-STD-810 unless otherwise specified.

4.8.2 Alignment for Normal Operation.

a. Apply power to equipment.

b. Align the power supply voltages per 3.4.11.2.

c. Align audio channels for 4 wire operation (-4 dbm in and -4 dbm out) using internal test tone oscillator and test align meter in accordance with TM 11-5805-383-12.

4.8.3 Standard Test Equipment. Standard test equipment is indicated in test figures by manufacturer and model number. In all cases, equivalent substitutes may be used. The equipment indicated in the figures is listed in Table VII. Standard test equipment shall be provided by the contractor. Government Technical Representative approval is required prior to use of proposed substitute items.

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TABLE VII - STANDARD TEST EQUIPMENT

Designation In Figure	Function	Manufacturer
Boonton 93AD	True RMS & DBM Digital Voltmeter	Boonton Electronics
Bowmar 462A	Transmission & Delay Measuring Set	Bowmar Instrument Corporation
Daven 12B	Transmission Measuring Set	Daven Division, Thomas A. Edison Industries
Fluke 8200A	AC/DC Digital Voltmeter	Fluke, Inc.
HP 200DC	Wide Range Oscillator	Hewlett-Packard Co.
HP 302A	Wave Analyzer	Hewlett-Packard Co.
HP 334A	Distortion Analyzer	Hewlett-Packard Co.
HP 400EL	Vacuum Tube Voltmeter	Hewlett-Packard Co.
HP 5245L	Electronic Counter	Hewlett-Packard Co.
Powertron 2505	AC Power Supply	Industrial Test Equipment Co., Inc.
Sierra 125B-CR	Wave Analyzer	Philco-Ford Corp.
TEK R5403/D40	Dual Trace Oscilloscope	Tektronix, Inc.

4.8.4 Special Test Equipment. Special test equipment is identified by name in test figures. Unless otherwise specified, special test equipment shall be provided by the contractor subject to approval of the Government. A description of this equipment is as follows:

a. AUX Simulator (see Figure 36). This circuit simulates:

(1) The presence of an auxiliary unit in the test of signals into and out of the TO AUX connectors.

(2) A time delayed signal (from an auxiliary unit) which is used to test the signal characteristics of the PCM from AUX connector.

b. Full Modulation Test Panel (see Figure 35). This panel provides a digital code for 2 kHz sine wave to be inserted into Channel 1 of TD-660( )/G. This panel shall plug into the 11A3 or 11A24 slot and the 11A3 or 11A24 panel shall plug into the test panel.

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c. Crystal Controlled Oscillator (see Figure 9). This circuit shall consist of a crystal controlled oscillator whose output frequency can be varied from 576 kHz - 0.005% to 576 kHz + 0.005%. This circuit shall be capable of synchronizing a Multiplexer TD-660( )/G through the SYNC IN connection (see 3.4.5.1 and 3.4.5.2).

d. Ancillary Items. Ancillary items required for testing by this specification shall be furnished by the contractor in the quantities necessary and with cable lengths as required to accomplish the testing herein. The ancillary items shall be as follows:

Interconnecting Cables - the interconnecting cables used with the equipment are as follows:

(1) Cable Assembly, Power Electrical CX-11727/G. The cable assembly shall be constructed in accordance with and meet the requirements of SC-DL-525135.

(2) Cable Assembly, Radio Frequency CG-1040( )/U. The cable assembly shall be constructed in accordance with and meet the requirements of SC-DL-531012.

(3) Cable Assembly, Special Purpose, Electrical CX-7870( )/TCC. The cable assembly shall be constructed in accordance with and meet the requirements of SC-DL-531014.

e. The Government shall furnish Multiplexer TD-660( )/G in test set-ups for First Article Tests and the first month's production tests of TD-660( )/G. After the first month's production is fabricated, the successful bidder shall use TD-660( )/G of his own manufacture in all production test set-ups and shall return all GFE TD-660( )/G units and their spare parts to the Government.

f. Panel Assembly, Dummy Test. The extender panel assembly shall be constructed in accordance with and meet the requirements of SC-DL-524666.

g. Panel Assembly, Coder Adjustment. The panel assembly shall be constructed in accordance with and meet the requirements of SC-DL-524668.

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h. Extender Cable (Power Supply). The extender cable shall consist of a multi-conductor cable with an 11 pin connector at each end. It shall be constructed for insertion between the chassis and the plug-in power supply subassembly for maintenance and alignment. The cable shall not be less than 18 inches or more than 20 inches in length.

i. Bracket, Mounting. The mounting bracket shall be constructed in accordance with SM-C-524874.

4.8.5 Operational Test.

a. Connect equipment per Figure 16.

b. Align unit under test for 12 channel operation (unit under test is master).

c. Measure DC output voltages of the Power Supply.

d. Measure 1 kHz audio gain for all channels of unit under test.

e. Measure 1 kHz Signal-to-Noise ratio for all channels of unit under test.

f. Measure 1 kHz Signal-to-Total Distortion ratio for all channels of unit under test.

g. Measure transmit clock frequency of unit under test.

h. Check all service facility readings for proper indication (SWITCHES I, II, and III) on unit under test.

i. Talk and listen on all 12 channels.

j. Align for 6 channel operation.

k. Talk and listen on all 6 channels.

l. The requirements of 3.15 shall be met.

4.8.6 System Test (service condition/non-operating).

a. Connect equipment per Figure 33.

b. Align equipment for 12 channel operation.

c. Measure 1.1 kHz audio gain for all channels both directions.

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d. Measure channel signal-to-noise ratio for all channels both directions.

e. Measure 1.1 kHz signal-to-total distortion both directions.

f. Talk and listen on all channels.

g. Measure frequency at SYNC OUT XMTR connector of unit under test.

h. The requirements of 3.4.1.6, 3.4.1.8, 3.4.5.1, shall be met.

4.8.7 System test (service condition/operating).

a. Connect equipment per Figure 32.

b. Align equipment for 24 channel operation.

c. Measure 1.1 kHz audio gain of all channels on unit under test in both directions.

d. Measure channel signal-to-noise ratio of all channels on unit under test in both directions.

e. Measure 1.1 kHz signal-to-total distortion ratio for all channels of unit under test in both directions.

f. Measure power supply dc voltages of unit under test.

g. The requirements of 3.4.1.6, 3.4.1.8, and 3.4.11.2 shall be met.

4.8.8 System Test (service condition temperature).

a. Connect equipment per Figure 32.

b. Align equipment for 24 channel operation. Adjust the power supply voltage per 3.4.11.2 (this step may be performed only once for each mode of operation during entire temperature test cycle).

c. Measure 1.1 kHz audio gain for all channels of unit under test in both directions.

d. Measure channel signal-to-noise ratio for all channels of unit under test in both directions.

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e. Measure 1.1 kHz signal-to-total distortion ratio of unit under test for all channels in both directions.

f. Measure signal-to-noise plus crosstalk ratio from all channels of unit under test to preceding and succeeding channel, near end and far end, and same channel near end, and in both directions.

g. Measure frequency of internal test tone oscillator of unit under test.

h. Measure dc power supply voltages of unit under test.

i. Disable noise generator and record code level output of all channels of unit under test.

j. Connect external oscillator to SYNC IN of unit under test with frequency set for 576,000 Hz.

k. Talk and listen on all channels of 12 channel system containing unit under test.

l. Talk and listen on channels 1, 4, and 7 of 12 channel system containing unit under test with frequency set for 576029 Hz and repeat for 575971 Hz.

m. Measure frequency at SYNC OUT XMIT connector of unit under test.

n. The requirements of 3.4.1.5.2, 3.4.1.6, 3.4.1.7, 3.4.1.8, 3.4.3.1, 3.4.5.1, and 3.4.11.2 shall be met.

o. Connect equipment per Figure 33.

p. Align equipment for 6 channel operation.

q. Repeat c. through j. above.

r. Talk and listen on all channels of 6 channel system containing unit under test.

s. Repeat l. above for channels 1, 3, and 5 of 6 channel system.

t. Repeat m. above.

u. The requirements of n. above shall be met.

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v. Disconnect external oscillator from SYNC IN of unit under test and place unit under test in MASTER position.

w. Measure transmit clock frequency of unit under test.

x. Conduct Digital Signal Test, 4.8.18 at -25°F and at +125°F.

y. The requirements of 3.4.5.2 shall be met.

4.8.9 Signal-to-Noise Ratio.

a. Connect equipment per Figure 2.

b. Align equipment for 6 channel operation.

c. Measure channel signal-to-noise ratio for all channels in both directions.

d. Align equipment for 12 channel operation.

e. Repeat c. above.

f. The requirements of 3.4.1.6 shall be met.

4.8.10 Frequency Response Test.

a. Connect equipment per Figure 4.

b. Align equipment for 12 channel operation. The oscillator HP 200 CD shall be adjusted until the meter is set at -4 dbm.

c. Measure Channel 1 gain for 200, 300, 500, 1100, 2100, 3450, 3500, and 3640 Hz.

d. Measure Channel 1 gain for 200, 300, 1100, 2100, 3500, and 3640 Hz at -14 dbm.

e. Repeat c. and d. above for Channels 4 and 7.

f. The requirements of 3.4.1.1 shall be met.

4.8.11 Output Level Test.

a. Connect equipment per Figure 5.

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b. Align equipment for 12 channel/2-wire operation.

c. Increase AG adjustment of Channel 1 until distortion is visible on scope. Note output level.

d. Reduce level of AG control and note minimum output.

e. Repeat c. and d. above for Channels 4 and 7.

f. Align equipment for 12 channel, 4-wire operation.

g. Adjust AG hairline and check output level for Channel 1, 4, 7, and 12.

h. The requirements of 3.4.1.3 and 3.4.9 shall be met.

4.8.12 Signal-to-Total Distortion.

a. Connect equipment per Figure 3.

b. Align equipment for 6 channel operation.

c. Measure channel 1 signal-to-total distortion with -4 dbm input at 300, 1100, and 3500 Hz.

d. Repeat for channels 3, 5, and 6.

e. Align equipment for 12 channel operation.

f. Repeat c. above.

g. Repeat c. for channels 4.7, and 12.

h. Measure channel 1 signal-to-total distortion with -27 dbm input at 1.1kHz.

i. Repeat h. above for channels 4 and 7.

j. The requirements of 3.4.1.8 shall be met.

4.8.13 Digital Squelch Test.

a. Connect equipment per Figure 6.

b. Align equipment for 12 channel operation.

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c. Apply 1100 Hz test tone signal and listen to unit under test on channels 1, 4. 7, and 11.

d. Place unit under test in an alarm condition by placing the MASTER-SLAVE switch in SLAVE position.

e. Listen in on channels 1, 4. 7, and 11.

f. Requirements of paragraph 3.4.3.3 shall be met.

g. Place unit under test in frame condition by placing MASTER-SLAVE switch in MASTER position.

h. Listen to 1100 Hz test tone in channels 1, 4. 7, and 11.

i. The requirements of paragraph 3.4.3.3 shall be met.

4.8.14 Frame alarm test.

a. Connect equipment per Figure 7.

b. Align equipment for normal operation.

c. Place MASTER-SLAVE switch on unit under test in SLAVE position.

d. Note that the FRAME-ALARM lamp lights and audible alarm sounds.

e. Note delay in audible alarm activation.

f. Silence audible alarm using pushbutton switch.

g. Place MASTER-SLAVE switch on unit under test in MASTER position.

h. FRAME ALARM lamp shall be extinguished and the audible alarm shall sound.

i. The requirements of 3.4.4 shall be met.

4.8.15 Master Slave Test.

a. Connect equipment per Figure 8.

b. Align equipment for 24 channel operation.

c. Measure channel signal-to-noise ratio of all channels of unit under test.

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d. Interchange MASTER and SLAVE units.

e. Measure channel signal-to-noise ratio of all channels of unit under test.

f. The requirements of 3.4.1.6 and 3.4.4.1 shall be met.

4.8.16 SYNC IN Test.

a. Connect equipment per Figure 9.

b. Apply 576 kHz synchronizing signal to the SYNC IN connector.

c. Measure frequency at SYNC OUT XMTR connector.

d. Measure phasing between SYNC IN and 12 CH TIM OUT.

e. Repeat b. and c. for the frequency of 576 kHz +0.005%.

f. Repeat b. and c. for the frequency of 576 kHz -0.005%.

g. The requirements of 3.4.5.1 and 3.4.5.2 shall be met.

4.8.17 Separation Test.

a. Connect equipment per Figure 10.

b. Align equipment for 12 channel operation.

c. Set MODE switch on TD-754( )/G to 24 channel operation. (The TD-754( )/G output signal then consists of a 12 channel PCM signal interleaved with a 288 kHz square wave). There shall be no evidence of frame alarm.

d. Observe 288 kHz square wave on scope.

e. The requirements of 3.4.4.2 shall be met.

4.8.18 Digital Signal Test.

a. Connect equipment per Figure 11.

b. Align equipment for 6 channel operation.

c. Measure characteristics of PCM and timing signals.

d. Measure phasing of PCM and timing signals.

e. Align equipment for 12 channel operation.

f. Repeat steps c. and d. above.

g. The requirements of 3.4.5.2 and 3.4.6.2 shall be met.

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4.8.19 Channel Alignment Test.

a. Connect equipment per Figure 13.

b. Align equipment for 12 channel operation.

c. Align channel 1 audio gain (AG) to give hairline with 2W-4W switch in 4W position.

d. Note channel output.

e. Align channel output for hairline with switch in 2W position.

f. Note channel output.

g. Repeat for channels 4 and 7.

h. Using external oscillator set for -4dbm input, align channel 1 audio gain for -4dbm output.

i. Apply internal test tone to channel one and note channel 1 output.

j. Connect equipment per Figure 14.

k. Apply internal test tone to channel one.

l. Measure frequency and percent distortion of test tone oscillator.

m. The requirements of 3.4.10 shall be met.

4.8.20 TO AUX Test and FROM AUX Test.

a. Connect equipment per Figure 12.

b. Align equipment for 6 channel operation.

c. Operate AUX IN-OUT switch to IN position.

d. Measure characteristics of AUX ADDRESS, RCVR AUX CLOCK, RCVR AUX ADDRESS, and PCM TO AUX signals.

e. Check for proper multiplexer operation for compliance with 3.4.12 by monitoring internal test tone on channel 1.

f. Align equipment for 12 channel operation.

g. Repeat c., d., and e. above.

h. The requirements of 3.4.10 and 3.4.12.1 through 3.4.12.4 shall be met.

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i. Connect equipment per Figure 15.

j. Align equipment for 12 channel operation.

k. Operate AUX IN-OUT switch to IN position.

l. Adjust AUX simulator such that PCM from AUX lags RCVR AUX CLOCK by 300 ns.

m. Apply a -4 dbm,1100 Hz signal to channels 9, 10, 11, and 12, (channels 5 and 6 for 6 channel operation).

n. Measure the Signal-to-Total Distortion Level on channels 9, 10, 11, and 12 (channels 5 and 6 for 6 channel operation).

o. Measure the Signal-to-Noise ratio on channels 1, 2, 3, and 4 (channels 1 and 2 for 6 channel operation).

p. Adjust AUX simulator such that PCM FROM AUX lags RCVR AUX CLOCK by 980 ns.

q. Repeat m. through o. above.

r. Align equipment for 6 channel operation.

s. Repeat l. through q. above.

t. The requirements of 3.4.1.6, 3.4.1.8, and 3.4.12.5 shall be met.

4.8.21 Input Level Test.

a. Connect equipment per Figure 5.

b. Insert full modulation test panel in 11A3/11A24 panel slot. Insert 11A3/11A24 panel into full modulation test panel. Align equipment for 12 channel operation with MASTER/SLAVE switch in MASTER position. See Figure 35 for the schematic diagram of the full modulation test panel.

c. Place switch on FULL MOD test panel to ON. Measure output level of channel 1.

d. Place switch on FULL MOD test panel to OFF. Insert a 1.1 kHz tone at channel 1 input. Increase the 1.1 kHz tone level until the output level of step c. above is attained.

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e. Measure the input level at this point. This is the full modulation level.

f. The requirements of 3.4.1.2 and 3.4.9 shall be met.

4.8.22 Input-Output Linearity Test.

a. Connect equipment per Figure 5.

b. Align equipment for 12 channel operation.

c. Measure 1.1 kHz output level of channel 1 for input levels of -35.0, -30.0, -20.0, -10.0, and -4.0 dbm.

d. Repeat for channels 4 and 7.

e. The requirements of 3.4.1.4 shall be met.

4.8.23 Signal-to-Noise plus Crosstalk Ratio Test.

a. Connect equipment per Figure 17.

b. Align equipment for 6 channel operation.

c. Measure Signal-to-Noise plus Crosstalk Ratio from channel 1 and 3 to all other channels, near end and far end, both directions.

d. Measure signal-to-noise plus crosstalk ratio from other channels to preceding and following channels far end, both directions, and to preceding same, and following channel near end, both directions.

e. Repeat c. and d. for 12 channel operation.

f. The requirements of 3.4.1.7 shall be met.

4.8.24 Delay Distortion Test.

a. Connect equipment per Figure 18.

b. Align equipment for 12 channel operation.

c. Measure channel 1 transmission delay at 600 and 3200 Hz and at maximum and minimum points between 600 and 3200 Hz.

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d. Compute delay distortion.

e. Repeat for Channels 4 and 7.

f. The requirements of 3.4.1.9 shall be met.

4.8.25 Audio Impedance Test.

a. Connect equipment per Figure 19.

b. Align equipment for 12 channel operation.

c. Measure channel 1 input and output impedance at 300, 1100, and 3500 Hz.

d. Repeat for channels 4 and 7.

e. The requirements of 3.4.1.10 shall be met.

4.8.26 Balance to Ground Test.

a. Connect the equipment per Figure 20.

b. Align equipment for 12 channel operation.

c. Apply -4 dbm signal to test circuit.

d. Measure input and output balance to ground for channel 1 at 300, 1100, and 3500 Hz.

e. Repeat for channels 4 and 7.

f. The requirements of 3.4.1.11 shall be met.

4.8.27 Hum Level Test.

a. Connect equipment per Figure 23.

b. Align equipment for 12 Ch operation.

c. Apply -4 dbm, 1.1 kHz tone. Measure the PAM peak-to-peak voltage at Pin 41 on panel 11A25 with the scope.

d. Remove 1.1 kHz tone and scope. Measure level of 60 and 120 Hz components at the same point.

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e. Requirements of 3.4.3.1.2 shall be met.

4.8.28 Digital Impedance Test.

a. Connect equipment per Figure 26.

b. Measure input impedance at 10 kHz of PCM IN, TIM IN, and SYNC IN connectors.

c. The requirements of 3.4.8 shall be met.

4.8.29 Channel Monitoring Test.

a. Connect the equipment per Figure 29.

b. Align the equipment for 12 Ch - 4 wire operation. Set SI to off, SII to off, SIII to TALK, and SIV to Ch 1.

c. Adjust the input level to obtain 488 mv (-4 dbm) at the input to Channel 1.

d. Measure the input levels at the Talk Monitor Mic at 300, 500, 1000, 1500, 2000, 2500, 3000, and 3500 Hz, to determine the frequency response.

e. Connect the equipment per Figure 30.

f. Align equipment for 12 Ch - 4 wire operation. Set SI to off, SII to off, SIII to off, and SIV to Ch 1.

g. Apply 869 mv (+1 dbm) to output of Ch 1.

h. Measure the frequency response characteristics from the output of audio channel to the Talk Monitor Earphone output at 300, 500, 1000, 1500, 2000, 2500, 3000, and 3500 Hz.

i. The requirements of 3.4.9 shall be met.

4.8.30 Electrical Interchangeability Test.

a. Connect equipment per Figure 16.

b. Replace panels 11A23 through 11A29 with spare panels (only two 11A26 panels shall be replaced) in unit under test.

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c. Align unit for 12 Ch operation using only the internal service facilities.

d. Measure 1 kHz audio gain for all channels of unit under test.

e. Measure 1 kHz signal-to-noise ratio for all channels of unit under test.

f. Measure 1 kHz signal-to-total distortion ratio for all channels of unit under test.

g. Check all service facility readings for proper indication (SWITCHES I, II, and III).

h. The requirements of 3.4.1.6, 3.4.1.8, 3.4.7, 3.4.10, and 3.12 shall be met.

4.8.31 Audio Level Stability Test.

a. Connect equipment per Figure 4.

b. Align equipment for 12 channel operation.

c. Set input to 1 kHz at -4 dbm and note all channel output levels.

d. Operate equipment for 72 hours and in this condition noting output levels every 24 hours.

e. The requirements of 3.4.1.5.1 shall be met.

4.8.32 Compressor Characteristics Test.

a. Connect equipment per Figure 21.

b. Align equipment for 12 channel operation.

c. Vary dc voltage into coder and note the dc in versus the code level out. A compression ratio curve similar to Figure 34 shall be plotted.

d. Compute compression ratio.

e. The requirements of 3.4.2 shall be met.

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4.8.33 Code Level Stability Test.

a. Connect equipment per Figure 22.

b. Align equipment for 12 channel operation.

c. Disable noise generator.

d. Note code level for each channel (see Figure 34).

e. Reactivate noise generator.

f. Operate equipment at standard ambient conditions for 72 hours, noting code level each 24 hours using procedure c., d., and e. above.

g. The requirements of 3.4.3.1.1 shall be met.

4.8.34 Random Noise Source Test.

a. Connect equipment per Figure 25.

b. Measure spectral density of noise source from 20 kHz to 200 kHz using 250 Hz bandwidth meter at J4/11A25.

c. Check for operation of monitor function of service facilities.

d. Disable the noise generator by shorting J2 to J3/11A25. Observe the output at J4 and the service facility monitor.

e. The requirements of 3.4.3.2 shall be met.

4.8.35 Metering Test.

a. Connect equipment per Figure 27.

b. Align equipment for 6 channel operation.

c. Using service facility controls, determine proper operation of all circuits except power supply metering by performing the following:

(1) Go-no-go indications (green and yellow area). Disable signal being monitored and note meter indication, and note that meter indicates a disabled condition.

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(2) Critical adjustments (hairline). Adjust +12 volts for hairline indication and note level at available test point. Calculate the percent accuracy. (Note: +7 volts must also be correct.)

d. Align equipment for 12 channel operation

e. Perform c above.

f. Connect equipment per Figure 28.

g. Align equipment for 24 channel operation.

h. Perform c. above.

i. The requirements of 3.4.7 and 3.4.10 shall be met.

4.8.36 AC Input Test .

a. Connect equipment per Figure 31.

b. Adjust variable power supply for 60 Hz and 115 volts.

c. Align equipment for 12 channel operation.

d. Measure ac power.

e. Measure channel signal-to-noise ratio for all channels.

f. Measure 1 kHz signal-to-total distortion ratio for all channels.

g. Disable noise generator and note code level out for all channels.

h. Repeat d., e., f., and g. with power supply adjusted for 47.5 Hz 109 volts; 47.5 Hz 121 volts; 400 Hz 115 volts; 420 Hz 109 volts; and 420 Hz 121 volts.

i. The requirements of 3.4.11.1, 3.4.1.6, 3.4.1.8, 3.4.3.1 shall be met.

4.8.37 Power Supply Output Test.

a. Connect equipment per Figure 31.

b. Adjust variable power supply for 60 Hz 115 volts.

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c. Align equipment for 12 channel operation.

d. Measure dc power supply output voltage and ripple. Ripple and voltage measurements shall be made from the front panel jacks with the Power Supply 11A33 installed in the chassis.

e. Repeat d., with power supply adjusted for 47.5 Hz 109 volts; 47.5 Hz 121 volts; 400 Hz 115 volts; 420 Hz 109 volts, and 420 Hz 121 volts.

f. The requirements of 3.4.11.2 shall be met.

4.9 Electrical Subassembly Test (see 3.13). Multiplexer TD-660( )/G and plug-in subassemblies, including running spare panel assemblies, shall be tested in accordance with their respective test requirement specification drawings listed in 3.13.

4.10 Electromagnetic Compatibility. The equipment shall be tested for compliance to the requirements of MIL-STD-461, Appendix A, Table A-1, for Class 1C equipment. The test methods and procedures described in MIL-STD-462 shall be employed. All electromagnetic compatibility tests shall be performed with the equipment simulating normal operation. The frequency range of the following tests are modified as follows: REO2 -14 kHz to 1000 MHz and RS03 -14 kHz to 1000 Mhz.

a. Connect equipment per Figure 24.

b. Align equipment for 12 channel operation.

c. Perform radiated and conducted emission tests on the equipment in accordance with MIL-STD-462 and the approved EMC/EMI test plan.

d. Perform radiated and conducted susceptibility tests on the equipment in accordance with MIL-STD-462 and the approved EMC/EMI test plan while monitoring the following:

(1) Signal-to-noise ratio on one audio channel.

(2) Signal-to-total distortion ratio on one audio channel.

(3) Listen on one audio channel.

e. The requirements of 3.11 shall be met.

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4.11 Inspection for Mechanical Interchangeability (see 3.12). The gage points identified on SC-GL-323304 shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.12.

4.12 Visual and Mechanical Inspection. Visual and mechanical inspection shall be in accordance with MIL-STD-252 and shall meet the requirements of 3.16.

4.13 TEMPEST (see 3.7). The TD-660( )/G shall be tested to the TEMPEST procedures of MIL-T-55654.

4.14 Reliability.

4.14.1 Reliability Testing.

4.14.1.1 First Article Reliability Test. The First Article reliability inspection testing shall be in accordance with Test Plan IV of MIL-STD-781.

4.14.1.2 Quality Conformance (Group D)Reliability Test. The reliability test for Group D inspection shall be in accordance with Test Plan IV A of MIL-STD-781.

4.14.1.3 Reliability Test Conditions. The detail test conditions, procedures and methods for the test described in 4.14.1 and 4.14.2 above shall be in accordance with the approved Reliability Test Plan for Multiplexer TD-660( )/G specified else where in the contract.

4.14.2 Reliability Test Procedure.

4.14.2.1 Performance Parameters to be Measured. The tests specified in Table VI shall be performed on all units subjected to Reliability testing.

4.14.2.2 Environmental and Operational Stresses.

4.14.2.2.1 Test level B of MIL-STD-781 shall apply except as noted below:

a. Input Voltage Cycling specified in Test level B does not apply.

b. Equipment ON-OFF cycling per Test level B is revised to read "Equipment ON-OFF Cycling. The equipment shall be turned on and operated in the 12 channel mode for 23 hours and then turned off for one hour. The next 24 hour period shall be in the 6 channel mode as indicated in Figure 1 with 23 hours operating and 1 hour off. Time expended to perform the operational tests shall be considered part of the "ON" period of the "ON-OFF" cycle. For these tests, 6 channel and 12 channel modes shall be used

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alternately. At the end of each operating period the mode shall be changed. During each 24 hour operating period (6 or 12 channel mode), the Operational Test (6 or 12 channel mode) of 4.8.5 (Table VI) shall be performed during the twenty-third hour and the test data shall be recorded. The test measurements shall not extend beyond the twenty-third hour.

4.14.2.2.2 Vibration. During testing, the equipment shall be mounted on a vibration table, positioned in its normal operating plane with all covers in place. Vibration shall be in a plane perpendicular to 11A23 and 11A29 printed circuit boards. Covers may be removed as necessary during test periods, but must be replaced at the conclusion of the electrical test period.

4.14.2.3 Maintenance Measurements. Daily operator adjustments and preventive maintenance shall only be made during the one hour "OFF" period. The extent of the adjustments and preventive maintenance permitted is only that outlined for the operator in the organizational Technical Manual (TM) 11-5805-382-12. These adjustments shall not be considered failures.

4.14.2.4 Failure Reporting and Analysis. A failure reporting and analysis system shall be implemented per MIL-STD-781. Final determination of the failure relevancy and multiple and pattern failure categorization shall be the responsibility of the procuring activity.

4.15 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (see 6.2.b). Preservation and packaging shall be level A or C as specified.

5.1.1 Level A.

5.1.1.1 Cleaning. Multiplexer TD-660( )/G shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Multiplexer TD-660( )/G shall be dried in accordance with the applicable procedure of MIL-P-116.

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5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged method IC-1.

5.1.1.4.2 Fuses. Fuses shall be packaged method III as follows: Cushion the fuses by wrapping together in paperboard conforming to PPP-P-291, type I. Secure the wrap with gummed paper tape conforming to PPP-T-45, type III, garde A.

5.1.1.4.3 Lamps. Each lamp shall be individually packaged method III as follows: Cushion each lamp by wrapping in paperboard conforming to PPP-P-291, type I. Secure the wrap with gummed paper tape conforming to PPP-T-45, type III, grade A.

5.1.1.4.4 Consolidated package. Consolidate the items, packaged as specified in 5.1.1.4.2 and 5.1.1.4.3, within a close-fitting box conforming to MIL-B-43014, form I, stype II. Fill all voids with paperboard conforming to PPP-P-291, type I to prevent movement. Box closure shall be in accordance with the appendix of the box specification.

5.1.1.4.5 Multiplexer TD-660( )/G. Each Multiplexer TD-660( )/G shall be individually packaged method III as follows: Secure the tool adjustment within the snap fastener provided on the unit. Cushion the item on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place the cushioned item together with the consolidated package as specified in 5.1.1.4.4, within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Box closure shall be in accordance with the appendix of the box specification.

5.1.2 Level C. Multiplexer TD-660( )/G shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling, and limited in-transit storate.

5.2 Packing. Packing shall be level A, B, or C, as specified. Shipping containers for all levels shall be capable of stacking and supporting

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superimposed loads during shipment and storage without damaging the container(s) or its contents (see 6.2.b).

5.2.1 Level A . A quantity of Multiplexer TD-660( )/G, packaged as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, stype 2 or 3, class 3. Fiberboard boxes being placed directly in the shipping container shall be waterproofed with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.2 Level B. A quantity of Multiplexer TD-660( )/G, packaged as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, domestic type; PPP-B-621, style 4, class 1, or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto.

5.2.3 Level C. A quantity of Multiplexer TD-660( )/G, packaged as specified in 5.1, shall be packed in shipping containers that comply with Federal or Military specifications, in a manner that will afford adequate protection to the package and its contents against physical and environmental deterioration and damage, during limited shipment, handling, and in-transit storage.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

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6.1 Intended Use. Multiplexer TD-660( )/G operates in Army Area Communication Systems. Operation in unprotected (canvas covered) installations requires the use of heaters to maintain the -25°F minimum operating temperature of Multiplexer TD-660( )/G and associated equipment such as Multiplexer TD-754( )/G; Converter, Telephone Signal CV-1548( )/G; and Radio Set AN/GRC-103( )(V). Transmission between Multiplexers TD-660( )/G is by radio set or cable system with the associated equipment listed above. Cable transmission systems also require the use of Restorer, Pulse Form TD-206( )/G.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Levels of preservation, packaging, and packing (see Section 5).

c. When rough handling and function test are required.

d. Marking and shipping of samples.

e. Place of Final Inspection.

f. Technical literature, tools and running spare parts.

g. First Article inspection (see 3.1).

h. Length of cables (if any are required).

6.3 Definitions.

6.3.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.3.2 Exfoliation. Exfoliation is corrosion along the grain boundries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.3.3 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

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6.3.4 Microbial growth. Microbial growth is the growth of very minute organisims. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.3.5 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.3.6 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.4 Group C Inspection. Approval to ship may be withheld at the discretion of the Government pending the decision from the Contracting Officer on the adequacy of corrective action (see 4.5.3.3).

6.4.1 Initial Shipments. Initial shipments of end items will be contingent only upon successful completion of all Group C tests (see 4.5.3).

6.5 Location of Operational Inspection. It is desirable that the operational system test (4.8.5)be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: TD-660B/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.7 Color. The color chip furnished by the procuring agency will be color chip No. X24087, and may be obtained upon request to Commanding Officer, US Naval Supply Depot, 5801 Taber Avenue, Philadelphia, Pennsylvania 19120.

6.8 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance providions established in this specification and the contract.

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The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:

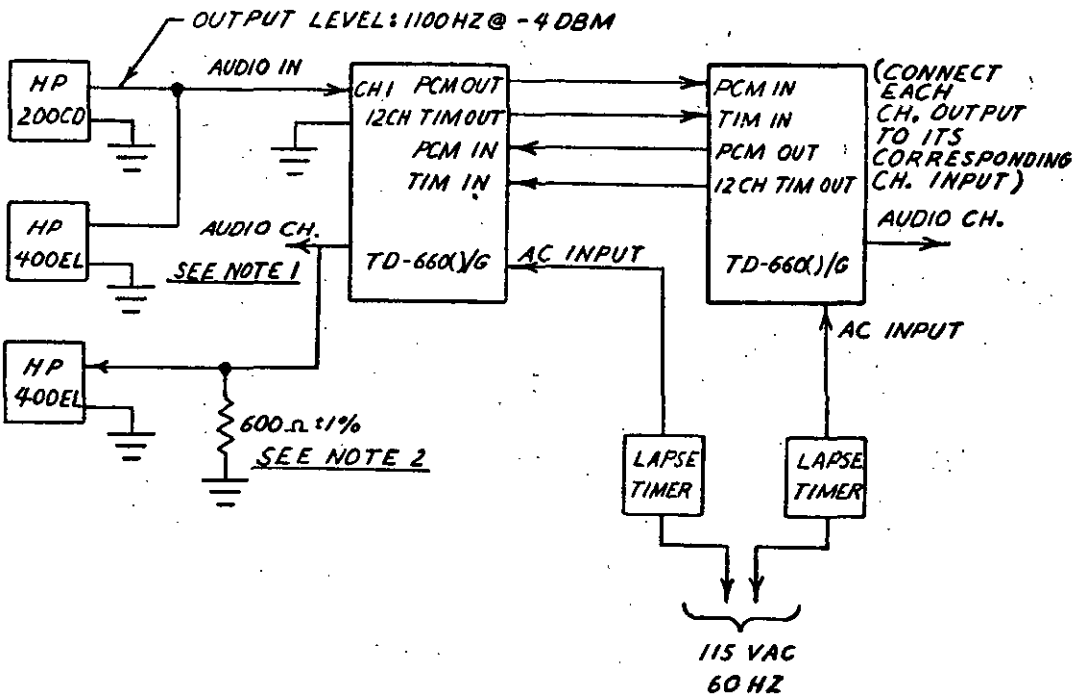
Army - (EL)

Preparing Activity

Army - (EL)

Project No. 5805-A227

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NOTES:

1. AUDIO CHANNEL INTERCONNECTIONS

   A. FOR 6 CH. OPERATION:

OUT	TO	IN
1	→	3
3	→	5
5	→	7
7	→	9
9	→	11

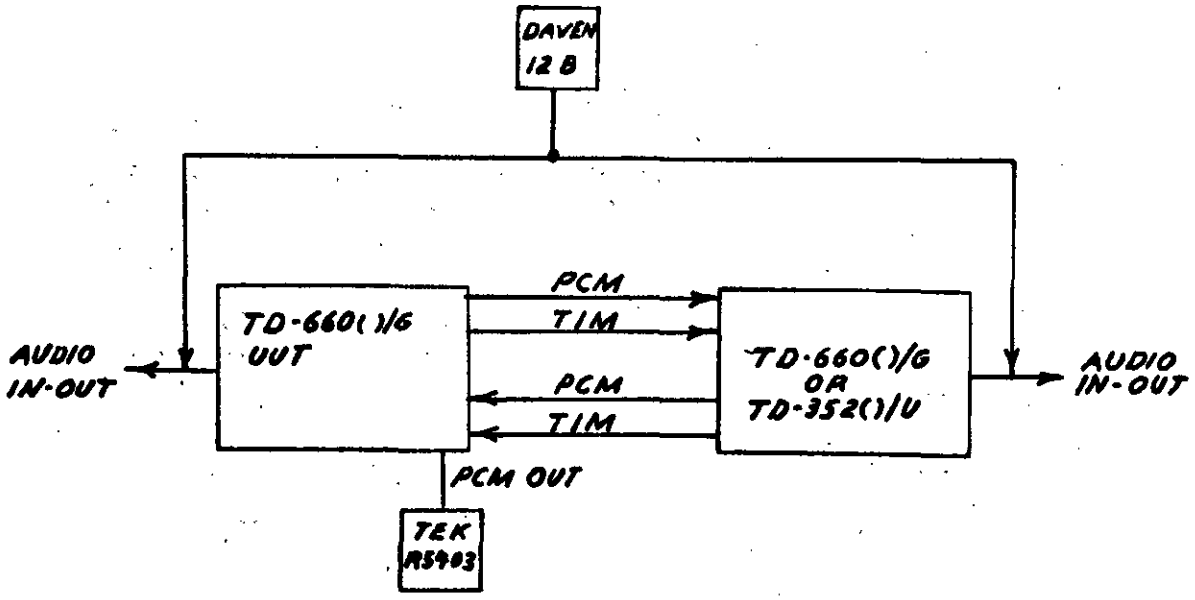
   B. FOR 12 CH. OPERATION, CONNECT AS FOR 6 CH. PLUS:

OUT	TO	IN
11	→	2
2	→	4
4	→	6
6	→	8
8	→	10
10	→	12

2. 600Ω LOAD CONNECTION, AUDIO OUT
   6 CH OPERATION, CH 11 OUT
   12 CH OPERATION, CH 12 OUT

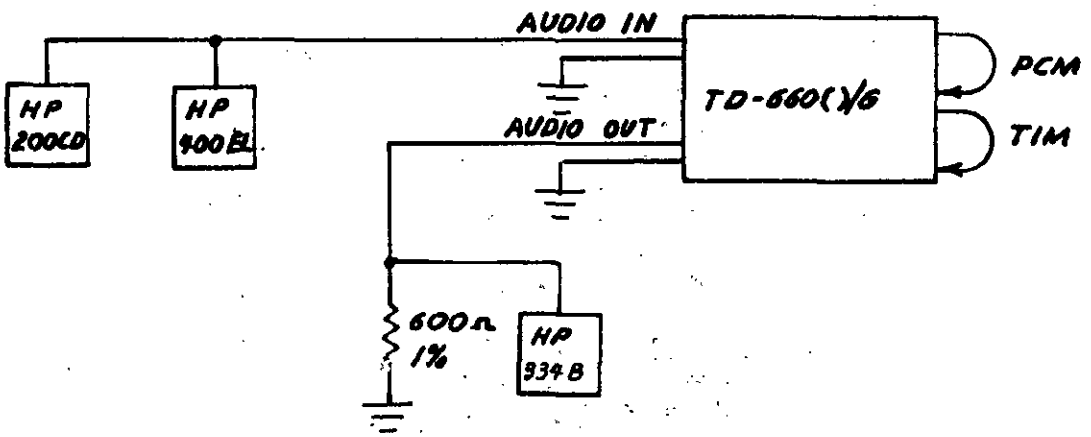
RELIABILITY/PRECONDITIONING TEST SET UP FOR MULTIPLEXER TD-660()/G
FIG. 1

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SIGNAL TO NOISE RATIO TEST

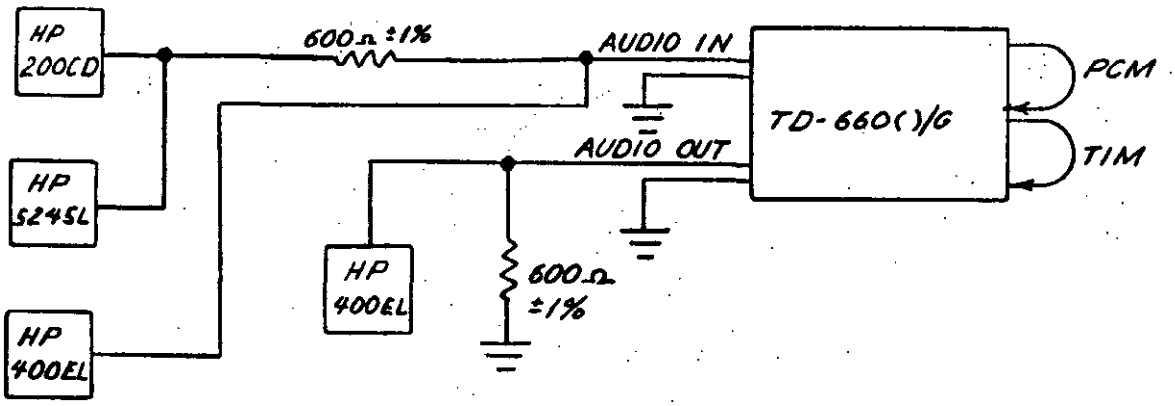
FIG. 2



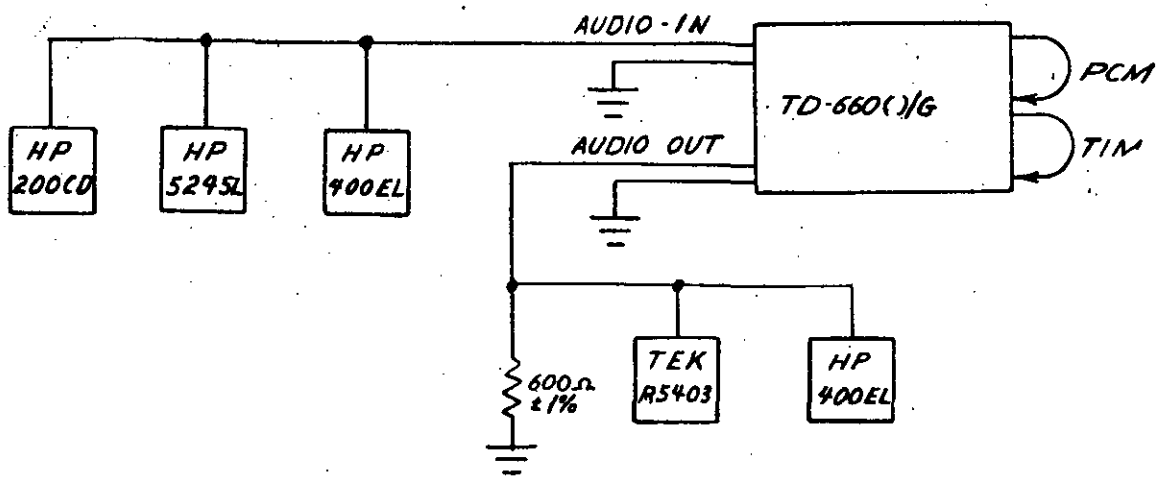
DISTORTION TEST

FIG. 3

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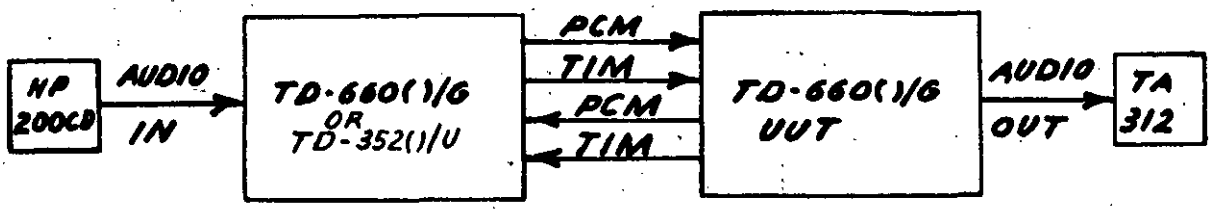


FREQUENCY RESPONSE TEST
FIG. 4

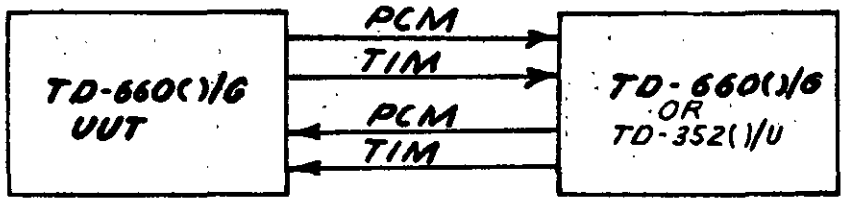


INPUT-OUTPUT LEVEL TEST
FIG. 5

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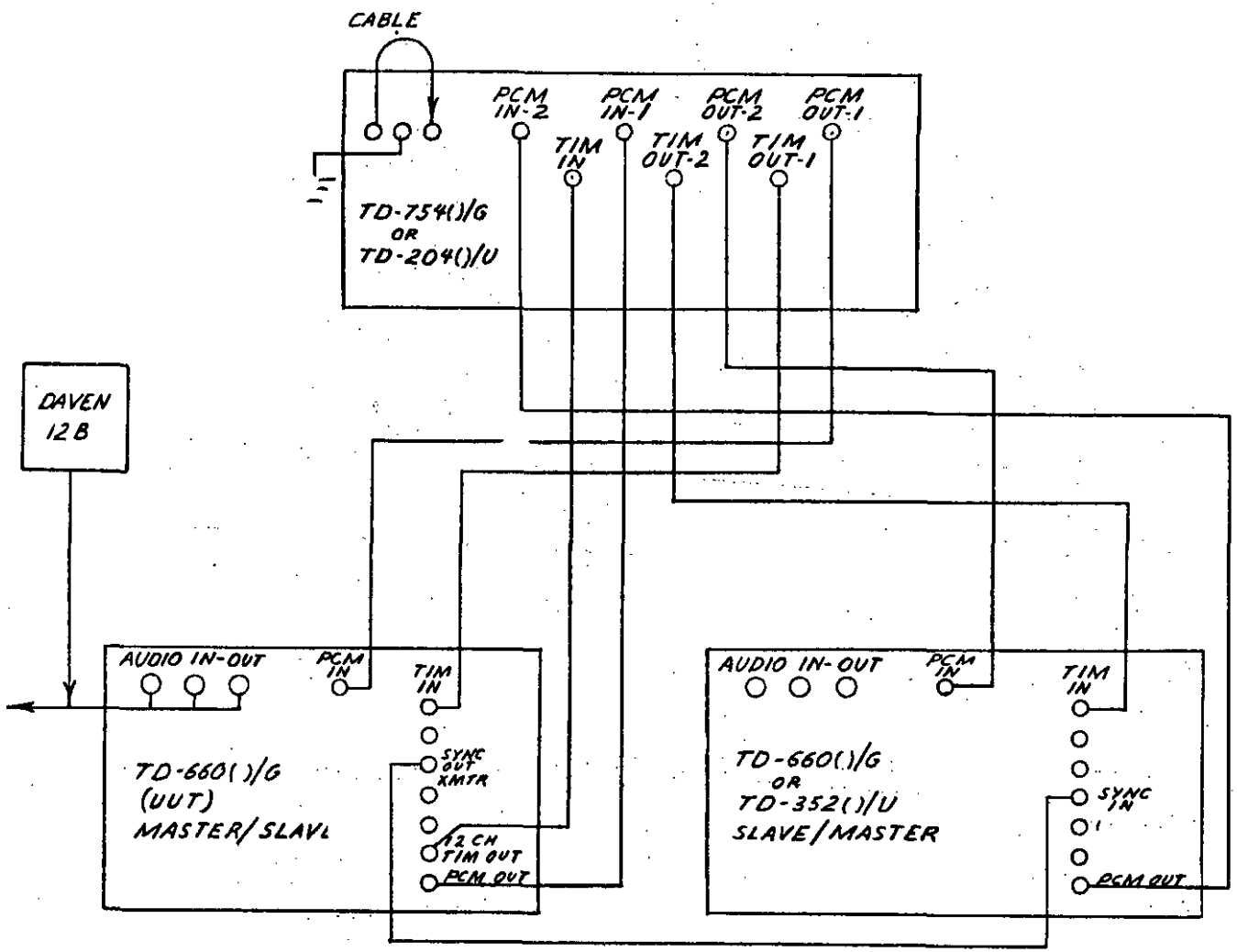


DIGITAL SQUELCH TEST
FIG. 6



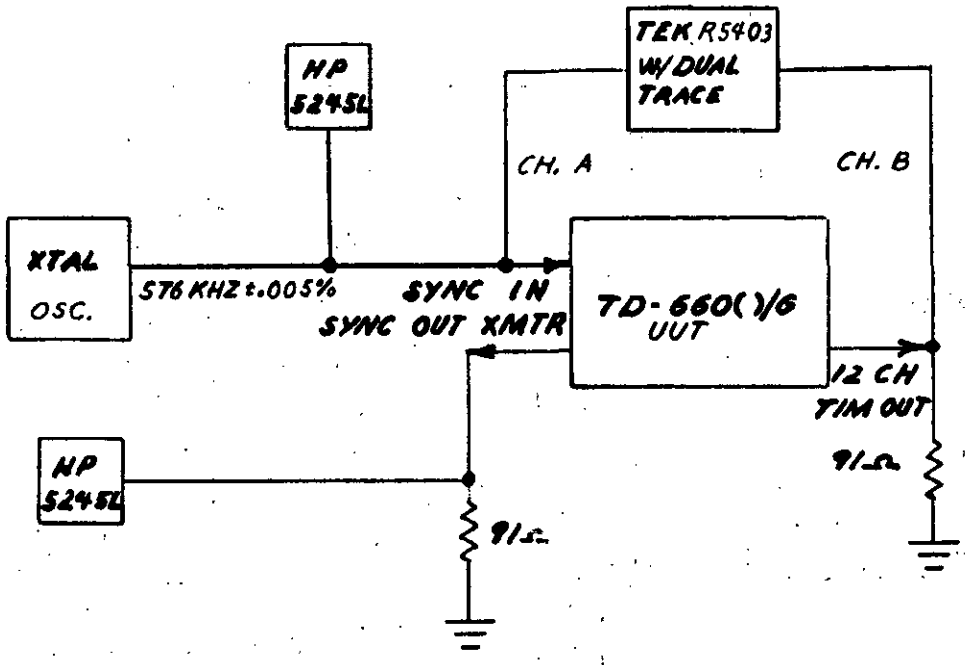
FRAME ALARM TEST
FIG. 7

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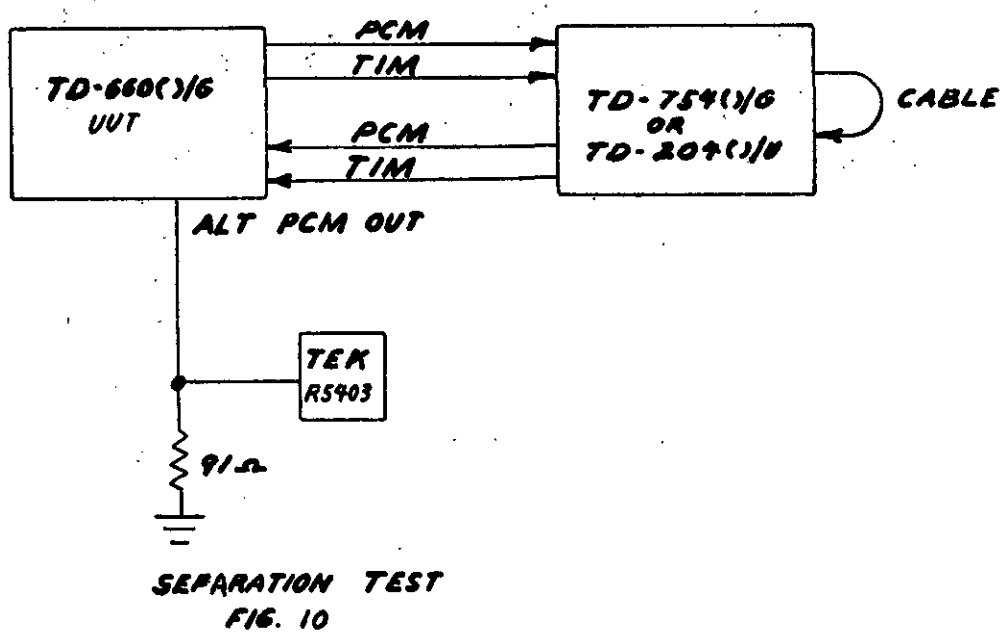


MASTER/SLAVE TEST
FIG. 8

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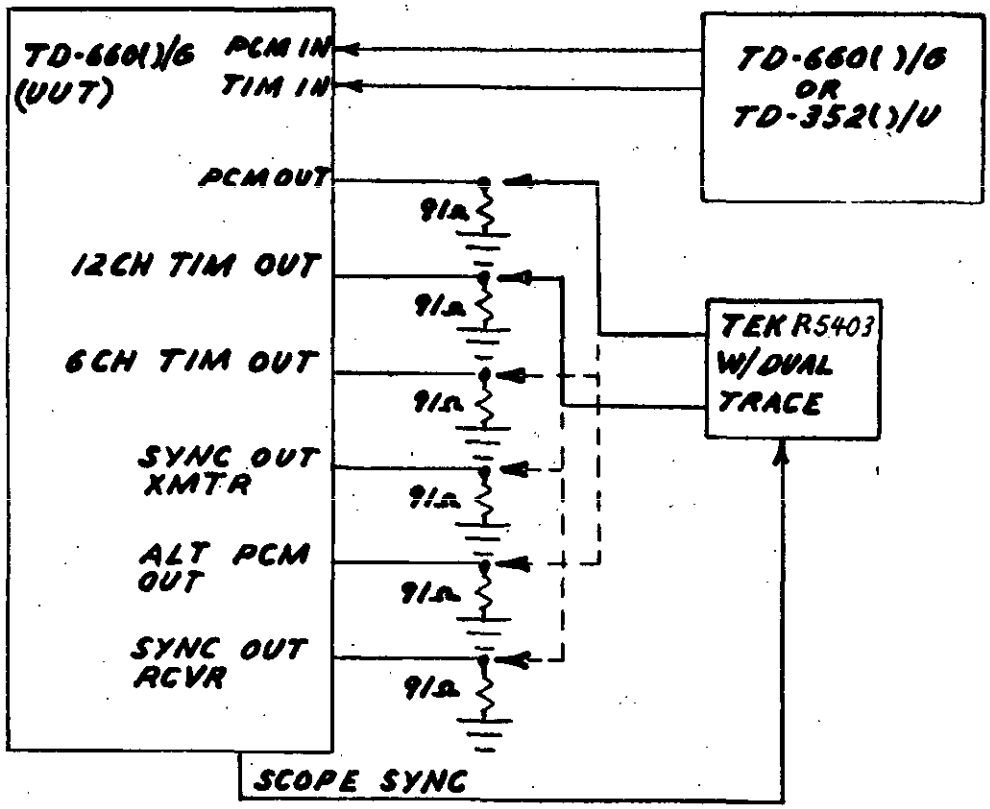


SYNC IN TEST
FIG. 9

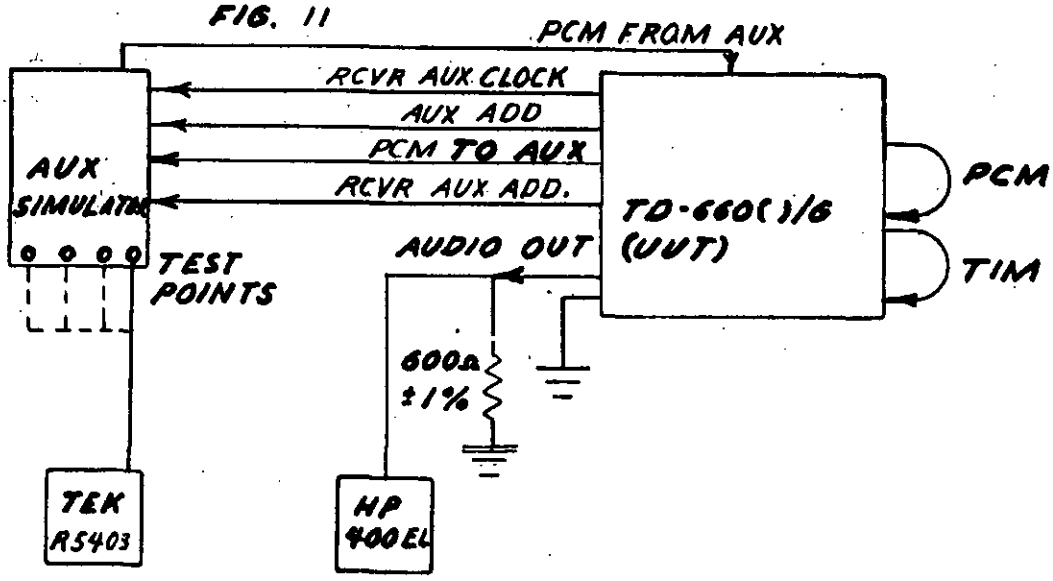


SEPARATION TEST
FIG. 10

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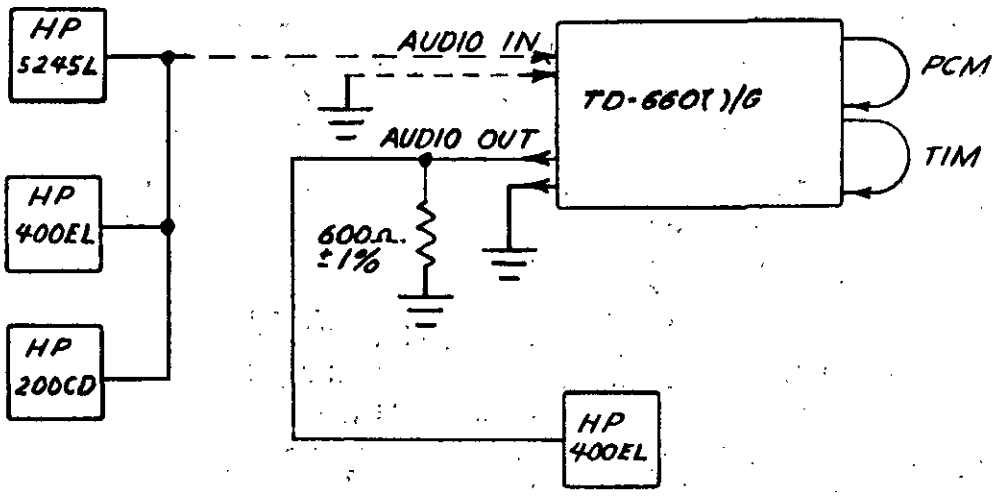


DIGITAL SIGNAL TEST
FIG. 11

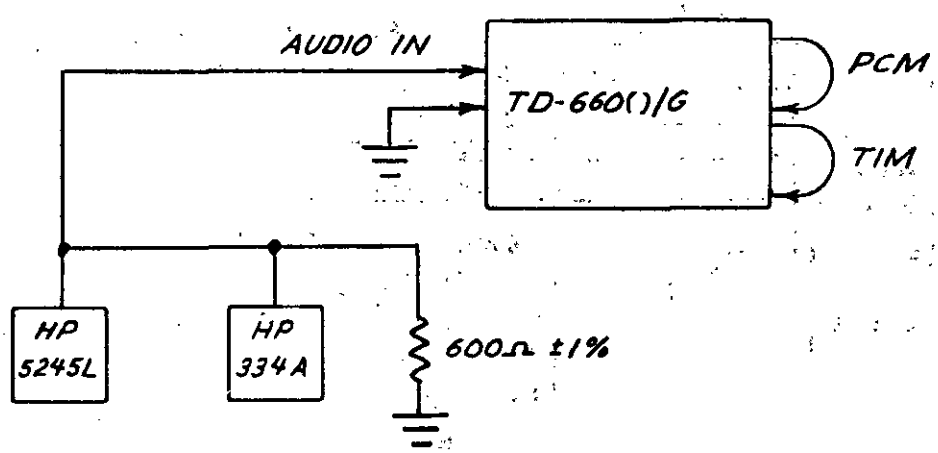


TO AUX TEST
FIG. 12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
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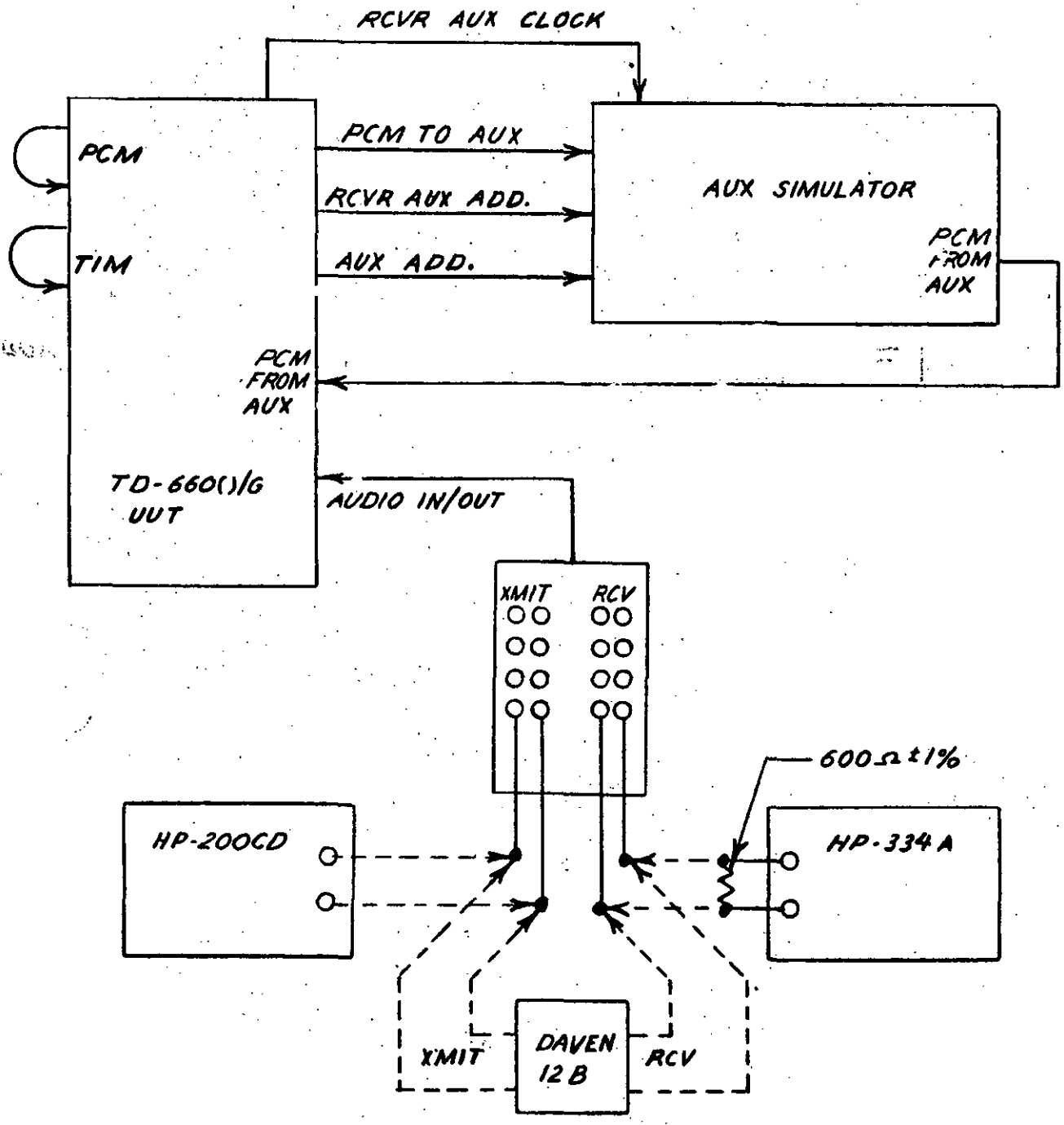


CHANNEL ALIGNMENT TEST
FIG. 13



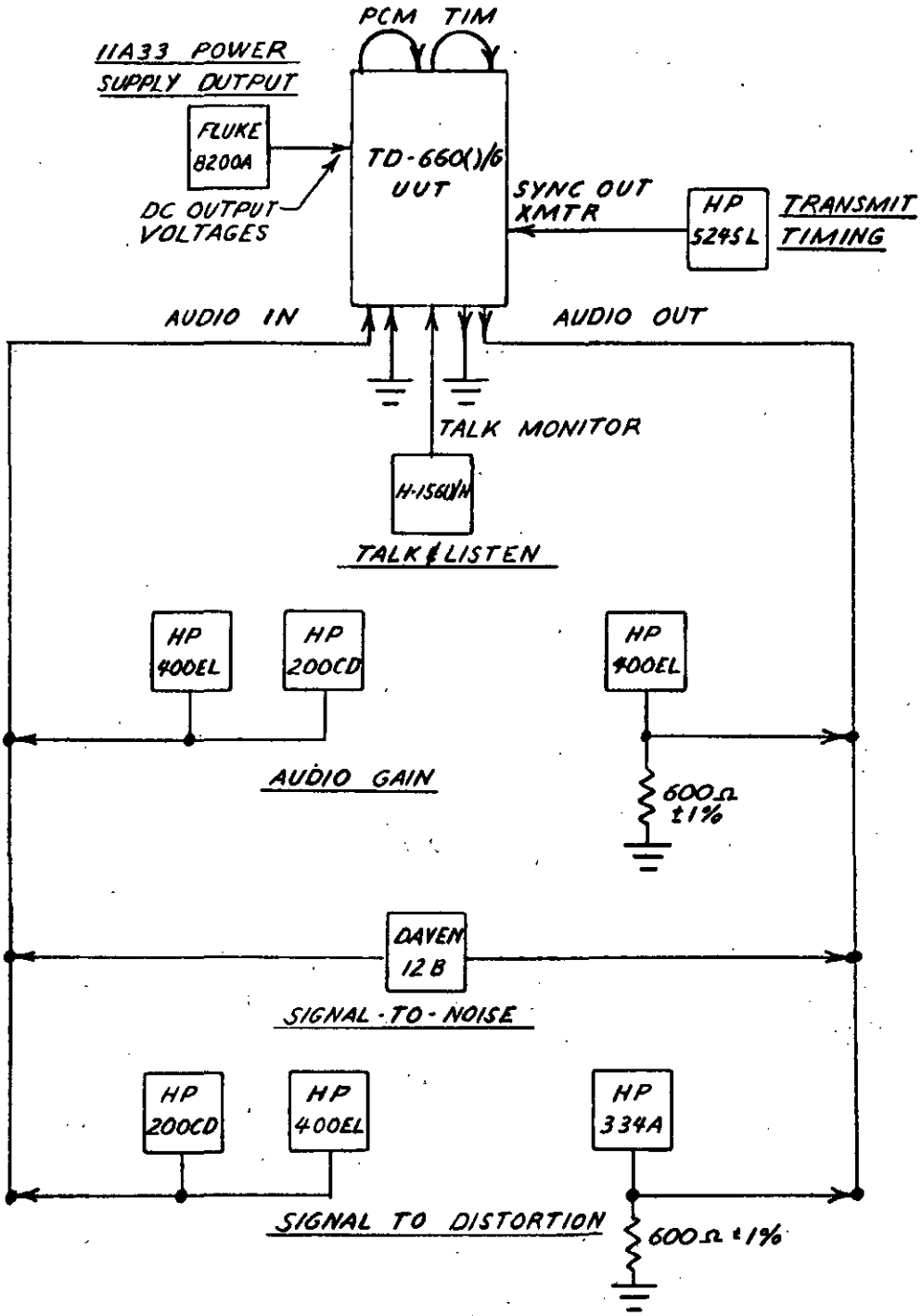
CHANNEL ALIGNMENT TEST
FIG. 14

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.



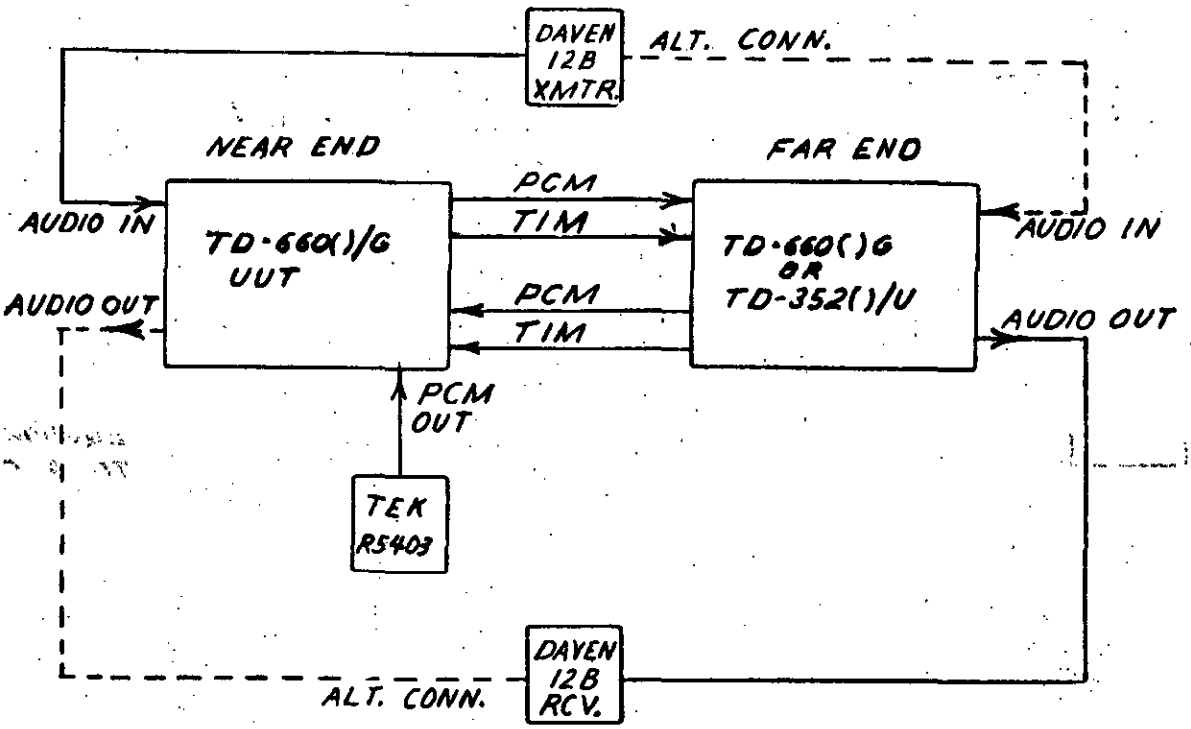
PCM FROM AUX TEST
FIG. 15

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
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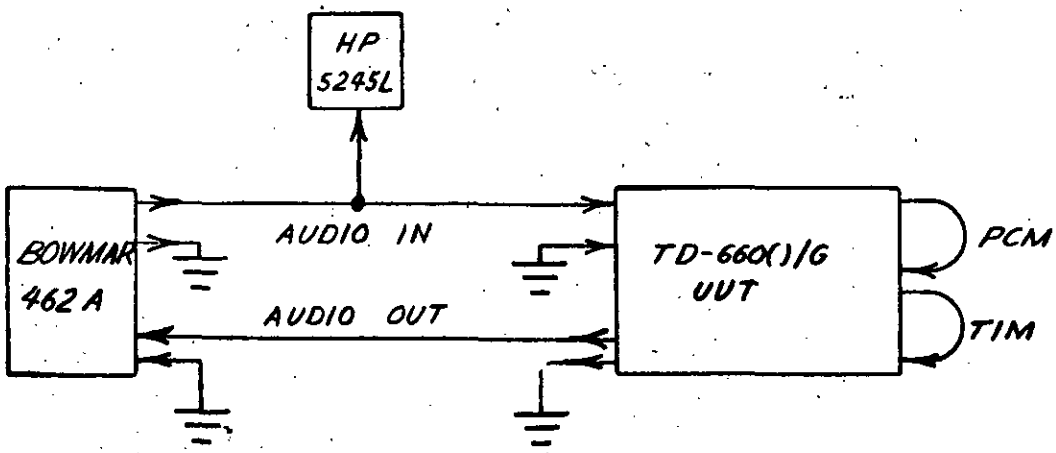


OPERATIONAL TEST
FIG. 16

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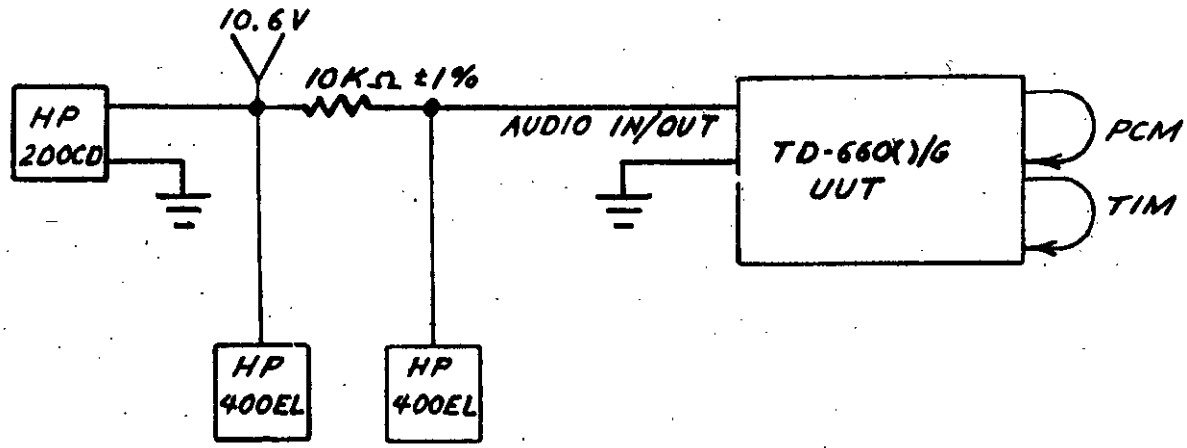


SIGNAL TO NOISE + CROSS TALK TEST
FIG. 17

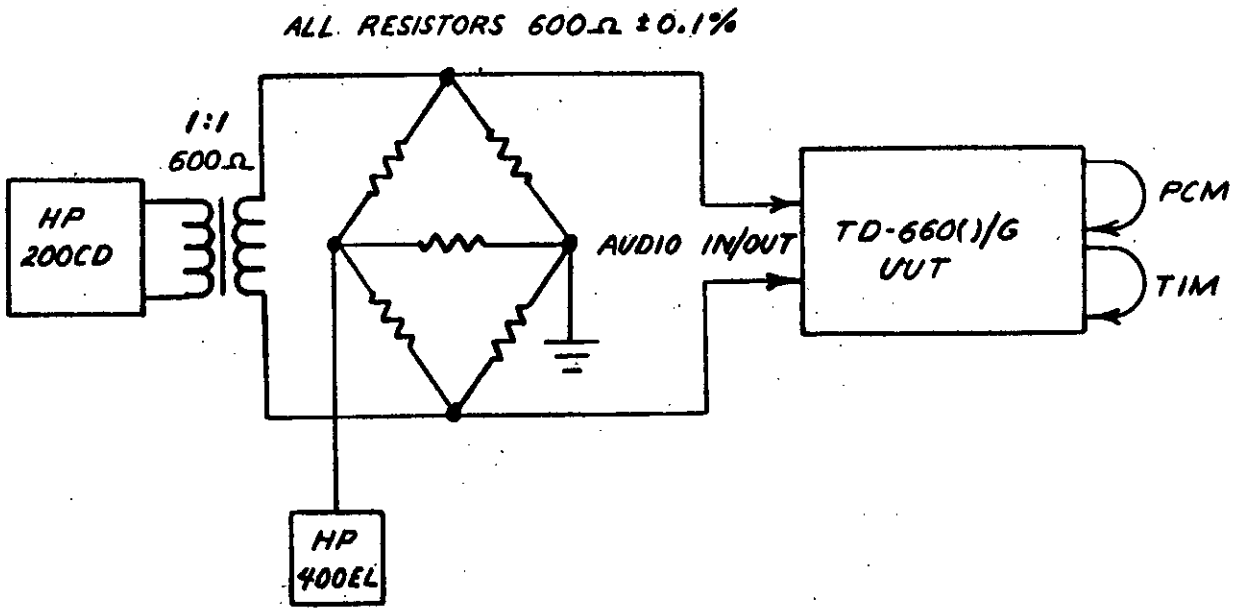


DELAY DISTORTION TEST
FIG. 18

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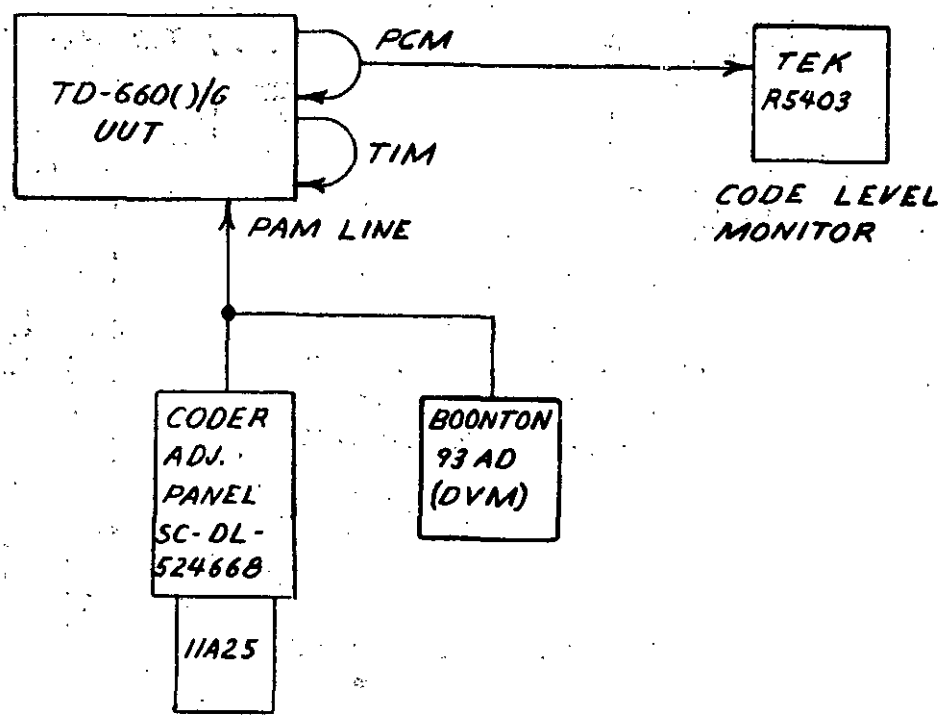


AUDIO IMPEDANCE TEST
FIG. 19

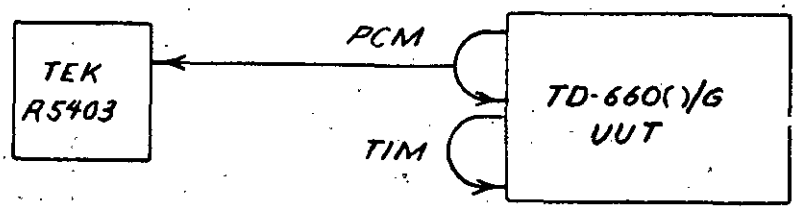


BALANCE TO GROUND TEST
FIG. 20

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
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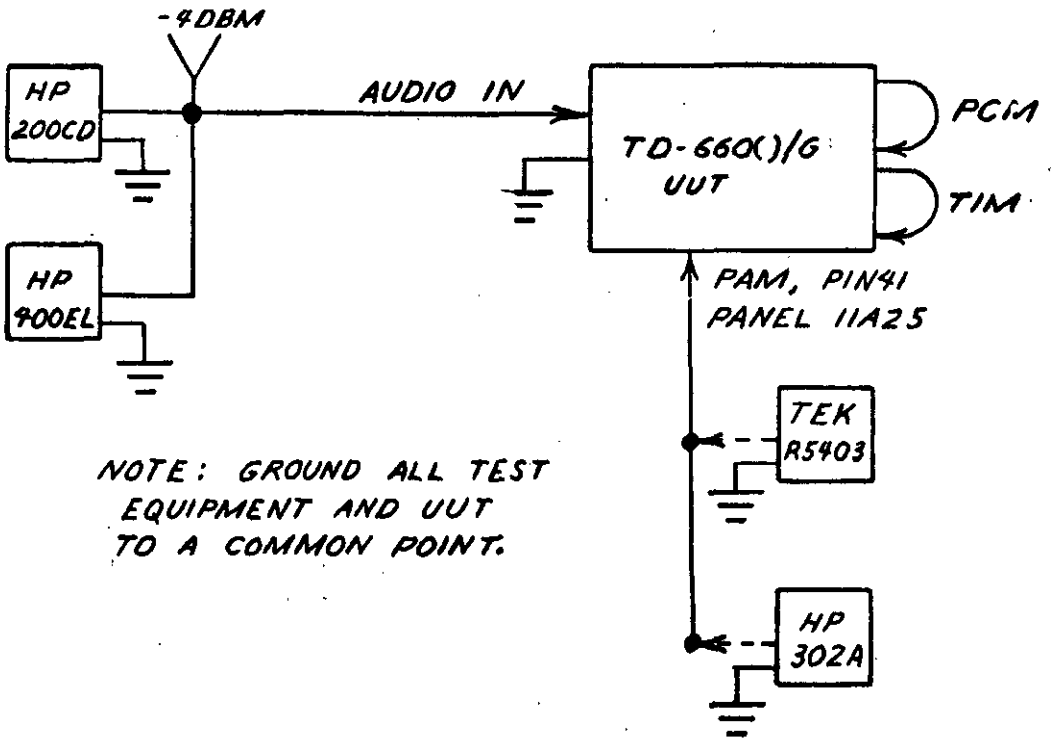


COMPRESSOR CHAR. TEST
FIG. 21

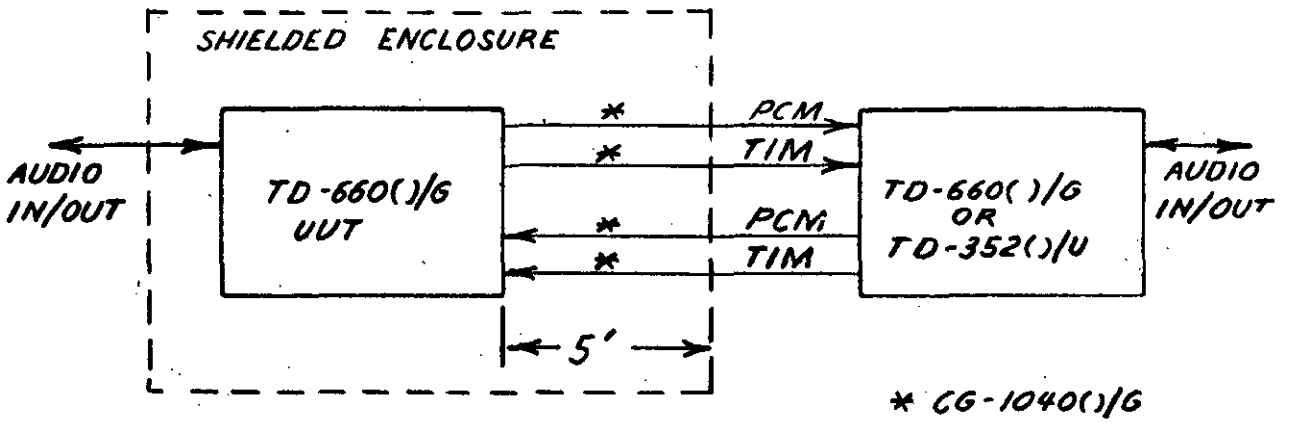


CODE LEVEL STABILITY TEST
FIG. 22

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

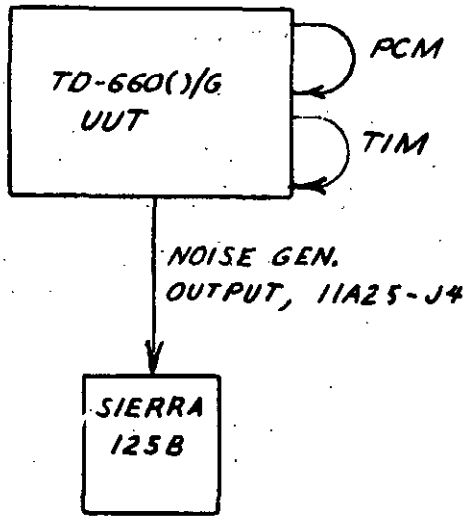


HUM LEVEL TEST
FIG. 23

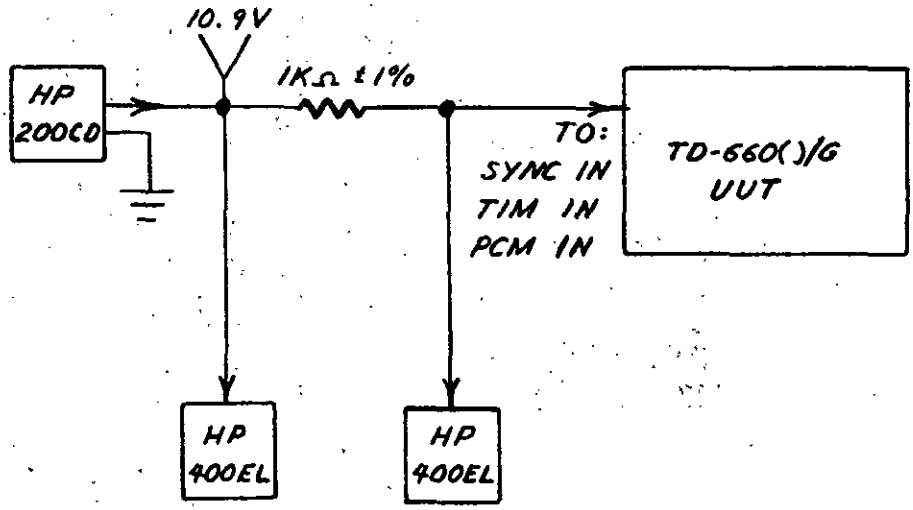


ELECTROMAGNETIC COMPATIBILITY TEST
FIG. 24

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

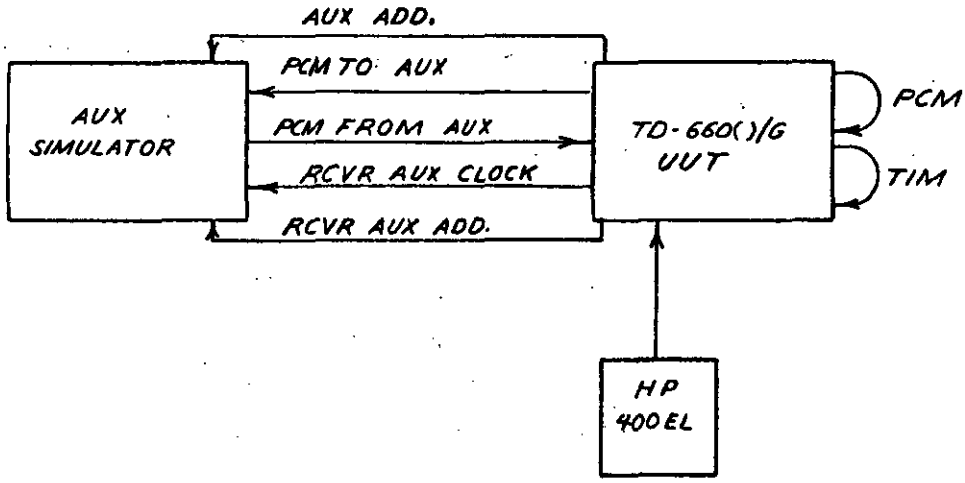


RANDOM NOISE SOURCE TEST
FIG. 25

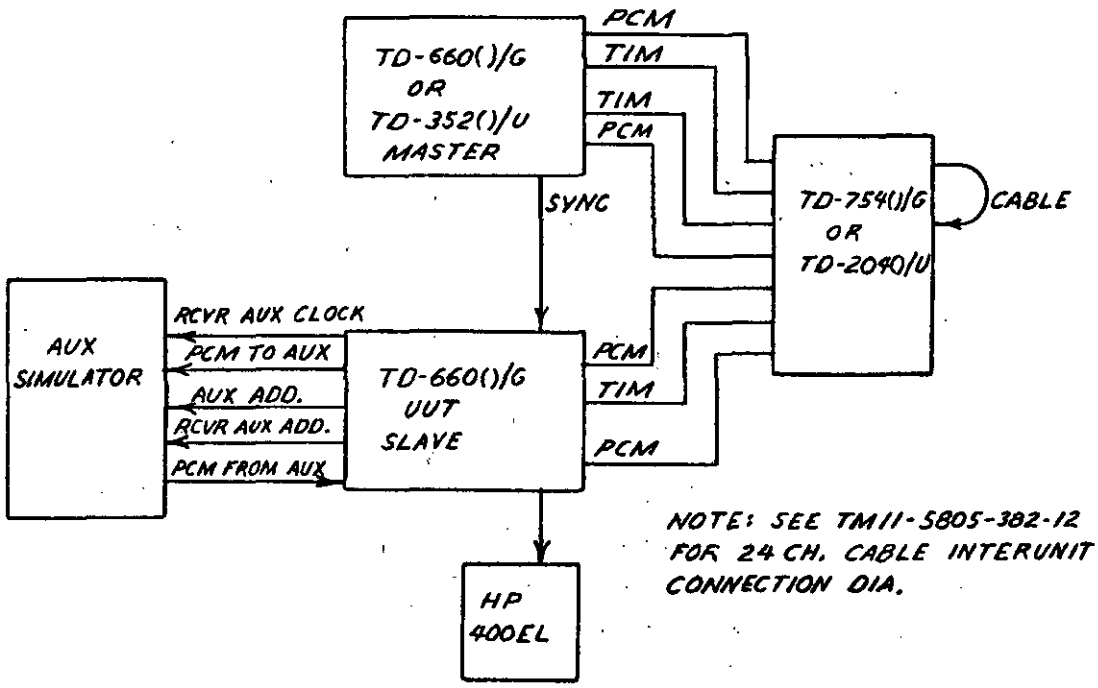


DIGITAL IMPEDANCE TEST
FIG. 26

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.



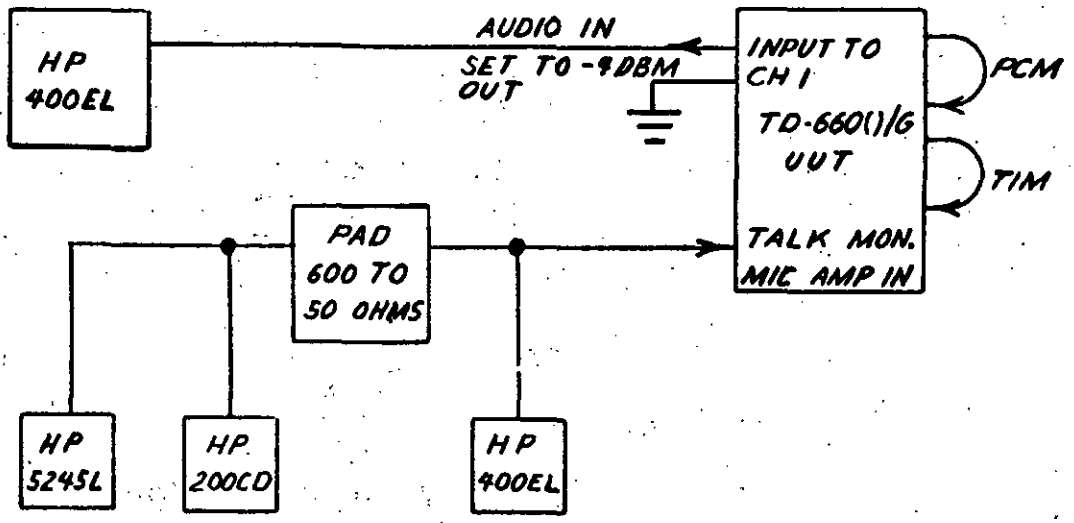
METERING 6/12 CHANNEL
FIG. 27



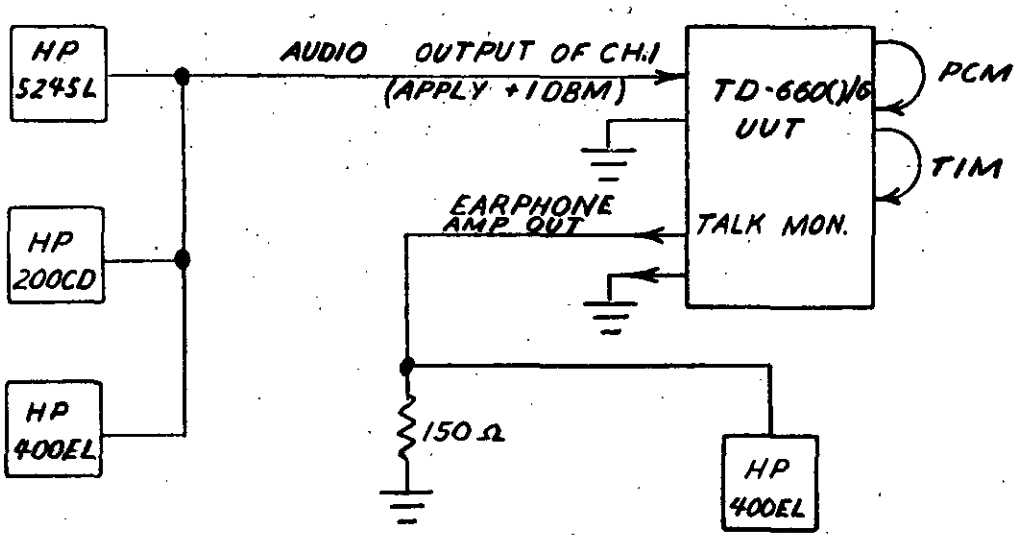
NOTE: SEE TM11-5805-382-12 FOR 24 CH. CABLE INTERUNIT CONNECTION DIA.

METERING 24 CHANNEL
FIG. 28

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
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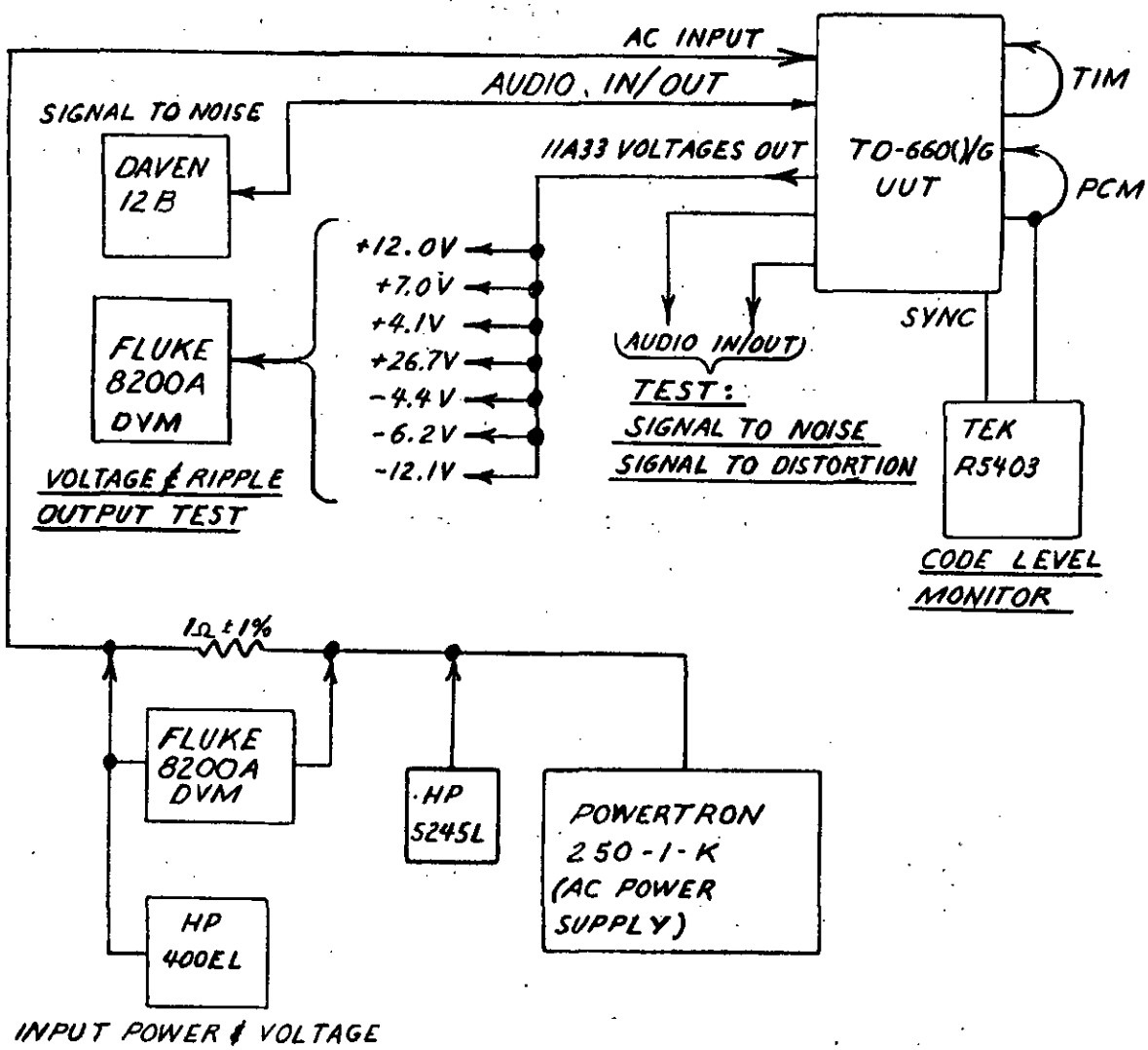


MICROPHONE AMPLIFIER TEST
FIG. 29



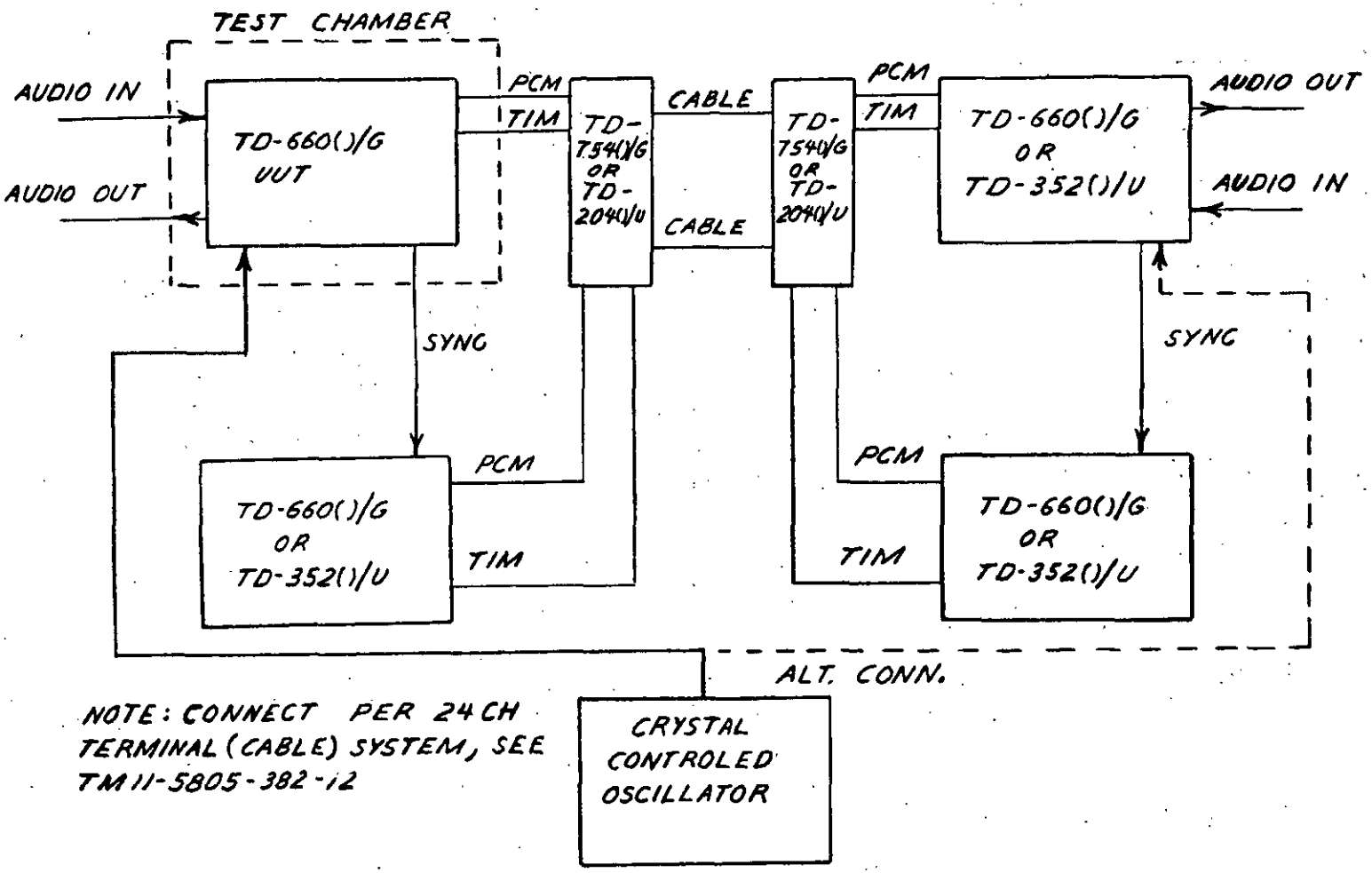
EARPHONE AMPLIFIER TEST
FIG. 30

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AC INPUT/OUTPUT TESTS
FIG. 31

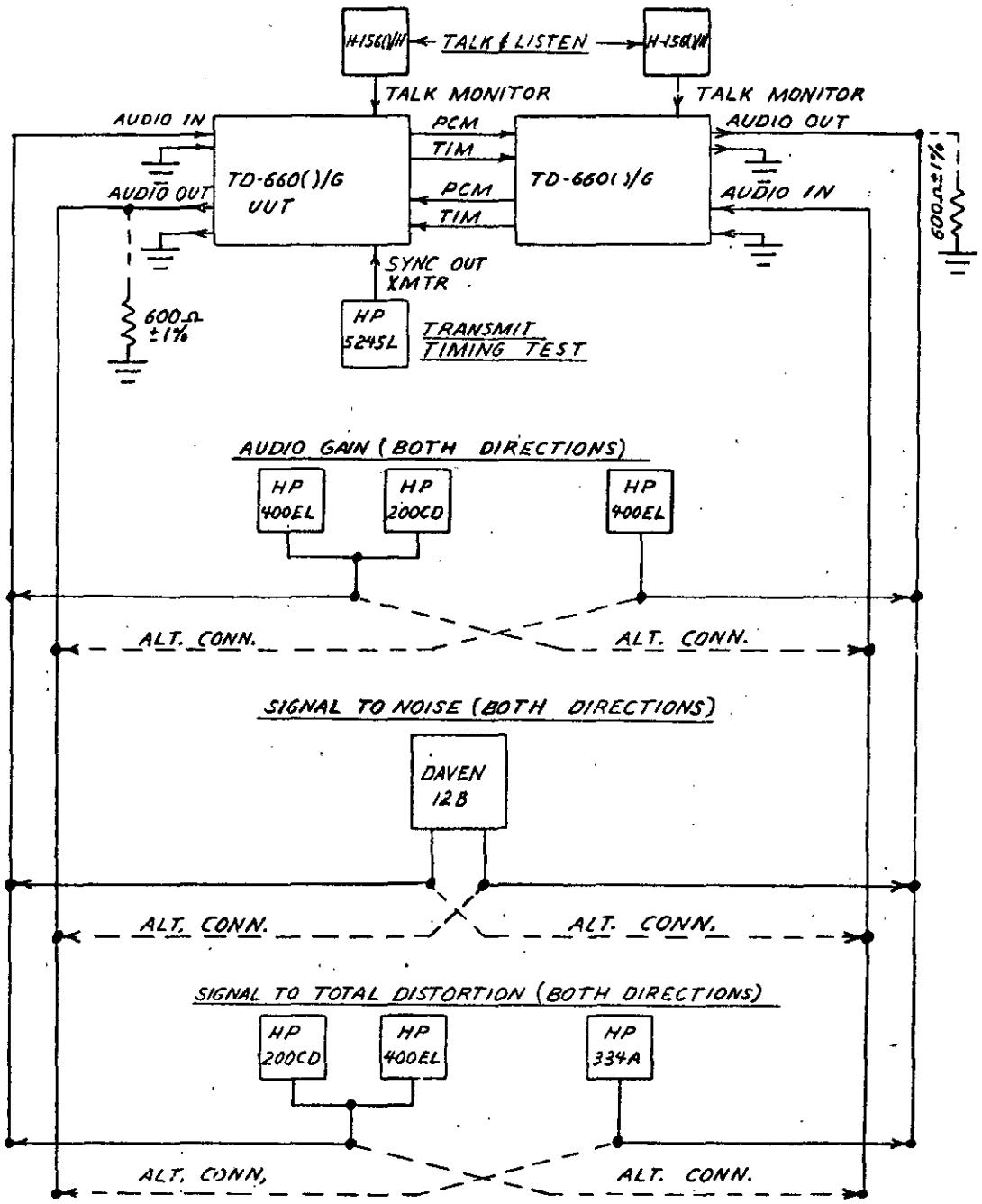
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.



NOTE: CONNECT PER 24 CH TERMINAL (CABLE) SYSTEM, SEE TM 11-5805-382-12

SYSTEM TEST (SERVICE CONDITION OPERATING)
FIG. 32

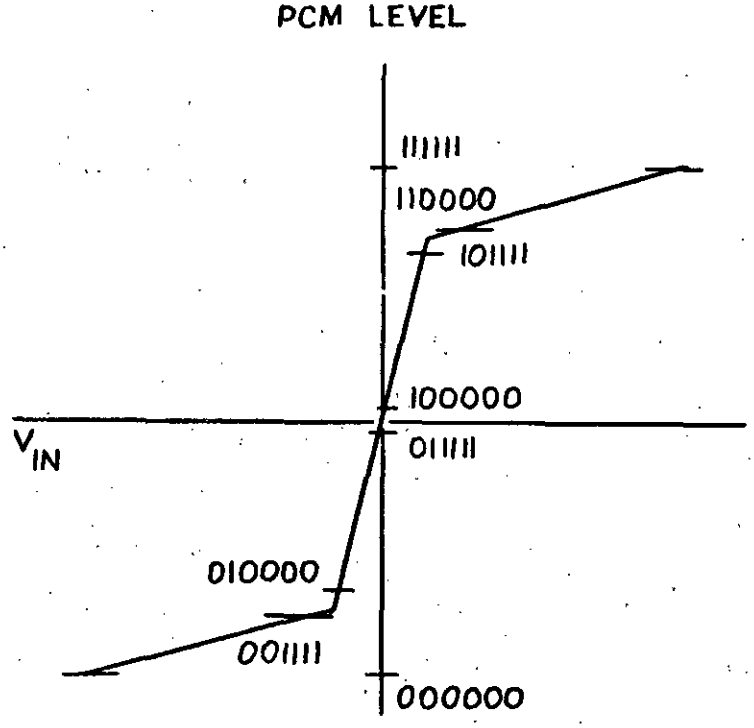
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SYSTEM TEST (SERVICE CONDITION NON-OPERATING)
FIG. 33

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CODER CHARACTERISTICS & COMPRESSION DEFINITION



COMPRESSION RATIO CALCULATION:

POSITIVE REGION;

$$\text{RATIO} = 20 \log \frac{M1}{M2}$$

$$M1 = \frac{17 \text{ (NO. OF CODE LEVELS FROM 32 TO 48)}}{\text{VOLTAGE @ CODE LEVEL 48} - \text{VOLTAGE @ CODE LEVEL 32}}$$

$$M2 = \frac{32 \text{ (NO. OF CODE LEVELS FROM 32 TO 63)}}{\text{VOLTAGE @ CODE LEVEL 63} - \text{VOLTAGE @ CODE LEVEL 32}}$$

NEGATIVE REGION;

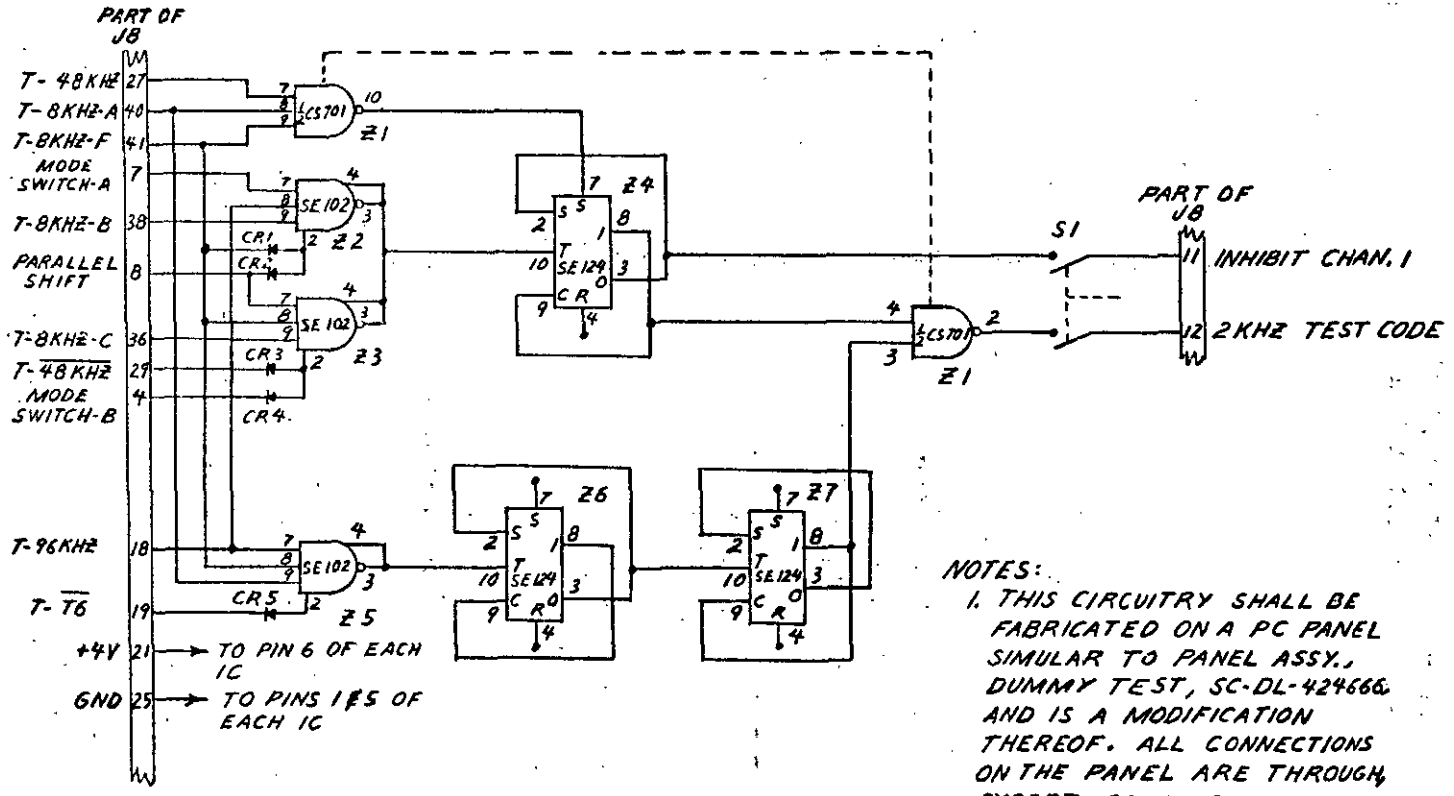
$$\text{RATIO} = 20 \log \frac{M1}{M2}$$

$$M1 = \frac{17 \text{ (NO. OF CODE LEVELS FROM 32 TO 16)}}{\text{VOLTAGE @ CODE LEVEL 32} - \text{VOLTAGE @ CODE LEVEL 16}}$$

$$M2 = \frac{32 \text{ (NO. OF CODE LEVELS FROM 32 TO 1)}}{\text{VOLTAGE @ CODE LEVEL 32} - \text{VOLTAGE @ CODE LEVEL 1}}$$

FIG. 34

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
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NOTES:

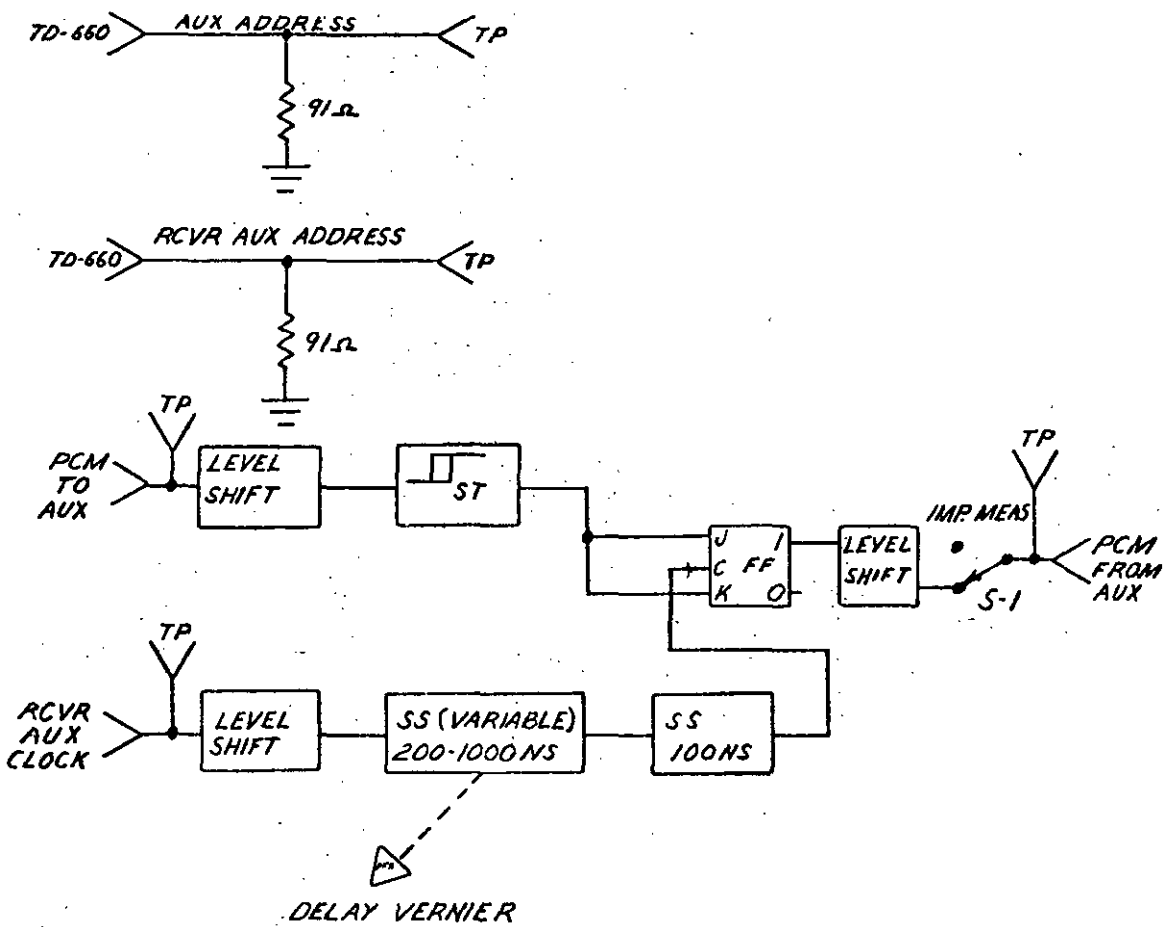
1. THIS CIRCUITRY SHALL BE FABRICATED ON A PC PANEL SIMULAR TO PANEL ASSY., DUMMY TEST, SC-DL-424666 AND IS A MODIFICATION THEREOF. ALL CONNECTIONS ON THE PANEL ARE THROUGH, EXCEPT CONTACTS 11 AND 12 ARE SWITCHED.

2. CR1, CR2, CR3, CR4, & CR5 ARE IN914 DIODES.

SCHEMATIC DIAGRAM, FULL MODULATION TEST PANEL ASSEMBLY

FIG. 35

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NOTE: LOAD IMPEDANCES AT INPUT TERMINATIONS AND SOURCE IMPEDANCES AT OUTPUT TERMINATIONS SHALL BE 91±10% OHMS. OUTPUT SIGNAL CHARACTERISTICS SHALL BE THE SAME AS PCM SIGNAL CHARACTERISTICS.

AUX SIMULATOR

FIG. 36

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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Commanding General
U.S.Army Electronics Command
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information, on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-M-55650B(EL) MULTIPLEXER TD-660( )/G

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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